LETTER

doi:10.1038/nature14184

Homologous-recombination-deficient tumours are dependent on Polθ-mediated repair

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Large-scale genomic studies have shown that half of epithelial ovarian cancers (EOCs) have alterations in genes regulating homologous recombination (HR) repair[1]. Loss of HR accounts for the genomic instability of EOCs and for their cellular hyper-dependence on alternative poly-ADP ribose polymerase (PARP)-mediated DNA repair mechanisms[2–5]. Previous studies have implicated the DNA polymerase θ (Polθ also known as POLQ, encoded by *POLQ*)[6] in a pathway required for the repair of DNA double-strand breaks[7–9], referred to as the error-prone microhomology-mediated end-joining (MMEJ) pathway[10–13]. Whether Polθ interacts with canonical DNA repair pathways to prevent genomic instability remains unknown. Here we report an inverse correlation between HR activity and Polθ expression in EOCs. Knockdown of Polθ in HR-proficient cells upregulates HR activity and RAD51 nucleofilament assembly, while knockdown of Polθ in HR-deficient EOCs enhances cell death. Consistent with these results, genetic inactivation of an HR gene (*Fancd2*) and *Polq* in mice results in embryonic lethality. Moreover, Polθ contains RAD51 binding motifs and it blocks RAD51-mediated recombination. Our results reveal a synthetic lethal relationship between the HR pathway and Polθ-mediated repair in EOCs, and identify Polθ as a novel druggable target for cancer therapy.

To examine changes in polymerase activity between tumours and normal tissues, we screened polymerase gene expression profiles in a large number of cancers (Supplementary Table 1). Gene set enrichment analysis (GSEA) revealed specific and recurrent overexpression of Polθ in EOCs (Extended Data Fig. 1a–c). Polθ was upregulated in a grade-dependent manner and its expression positively correlated with numerous mediators of HR (Extended Data Fig. 1d–j). As Polθ has been suggested to play a role in DNA repair[7–10], we investigated a potential role for Polθ in HR repair.

To test the relationship between Polθ expression and HR, we used a cell-based assay in human cells which measures the efficiency of recombination of two GFP alleles (DR-GFP assay)[14]. Knockdown of Polθ with short interfering RNA (siRNA) (Extended Data Fig. 2a) resulted in an increase in HR efficiency, similar to that observed by depleting the anti-recombinases PARI or BLM[15,16]. Depletion of Polθ caused a significant increase in basal and radiation (IR)-induced RAD51 foci (Fig. 1a, b and Extended Data Fig. 2b–d), and depletion of Polθ in 293T cells conferred cellular hypersensitivity to mitomycin C (MMC) and an increase in MMC-induced chromosomal aberrations (Extended Data Fig. 2e, f). These findings suggest that human Polθ inhibits HR and participates in the maintenance of genome stability.

Given that Polθ shares structural homology with coexpressed RAD51-binding ATPases (Extended Data Fig. 1k, l), we hypothesized that Polθ might regulate HR through an interaction with RAD51. RAD51 was detected in Flag-tagged Polθ immunoprecipitates, and purified full-length Flag–Polθ bound recombinant human RAD51 (Fig. 1c, d). Pull-down assays with recombinant GST–RAD51 and *in vitro* translated Polθ truncation mutants defined a region of Polθ binding to RAD51 spanning amino acids 847–894 (Fig. 1e, f and Extended Data Fig. 2g, h). Sequence homology of Polθ with the RAD51 binding domain of *C. elegans* RFS-1 (ref. 17) identified a second binding region (Extended Data Fig. 2i). Peptides arrays narrowed down the RAD51 binding activity of Polθ to three distinct motifs (Fig. 1g and Extended Data Fig. 2j). Substitution arrays confirmed the interaction and highlighted the importance of the 847–894 Polθ region as both necessary and sufficient for RAD51 binding (Extended Data Fig. 3a, b). Together these results indicate that Polθ is a RAD51-interacting protein that regulates HR.

In order to address the role of Polθ in HR regulation, we assessed the ability of wild-type or mutant Polθ to complement the siPolθ-dependent increase in RAD51 foci. Full-length wild-type Polθ fully reduced IR-induced RAD51 foci, unlike Polθ mutated at ATPase catalytic residues (A-dead) or Polθ lacking interaction with RAD51 (ΔRAD51) (Fig. 2a, b). Expression of a Polθ mutant lacking the polymerase domain (ΔPol1) was sufficient to decrease IR-induced RAD51 foci, suggesting that the N-terminal half of Polθ is sufficient to disrupt RAD51 foci (Fig. 2b and Extended Data Fig. 3c, d). We next measured the ability of wild-type or mutant Polθ to complement the siPolθ-dependent increase in HR efficiency. Again, expression of full-length Polθ or ΔPol1 decreased the recombination frequency when compared to cells expressing other Polθ constructs, suggesting that the N-terminal half of Polθ containing the RAD51 binding domain and the ATPase domain is needed to inhibit HR (Fig. 2c and Extended Data Fig. 3e).

A purified recombinant Polθ fragment (ΔPol2) from insect cells exhibited low levels of basal ATPase activity, as previously reported[18] (Fig. 2d, e). Polθ ATPase activity was selectively stimulated by the addition of single-stranded DNA (ssDNA) or fork DNA (Fig. 2e and Extended Data Fig. 4a). Electrophoretic mobility gel shift assays (EMSA) showed specific binding of Polθ to ssDNA (Fig. 2f and Extended Data Fig. 4b). We incubated ΔPol2 with ssDNA and measured RAD51–ssDNA nucleofilament assembly. Interestingly, RAD51–ssDNA assembly was reduced by ΔPol2 wild-type but not by A-dead or ΔRAD51, indicating that Polθ negatively affects RAD51–ssDNA assembly through its RAD51 binding and ATPase activities (Fig. 2g and Extended Data Fig. 4c–f). Furthermore, Polθ decreased the efficiency of D-loop formation, confirming that Polθ is a negative regulator of HR (Fig. 2h and Extended Data Fig. 4g–j).

As Polθ is upregulated in subgroups of cancers associated with HR deficiency (Fig. 3a) and Polθ activity shows specificity for replicative-stress-mediated structures (ssDNA and fork DNA) (Fig. 2e, f), we examined the cellular functions of Polθ under replicative stress. Subcellular fractionation revealed that Polθ is enriched in chromatin in response to ultraviolet (UV) light; and RAD51 binding by Polθ was enhanced by UV

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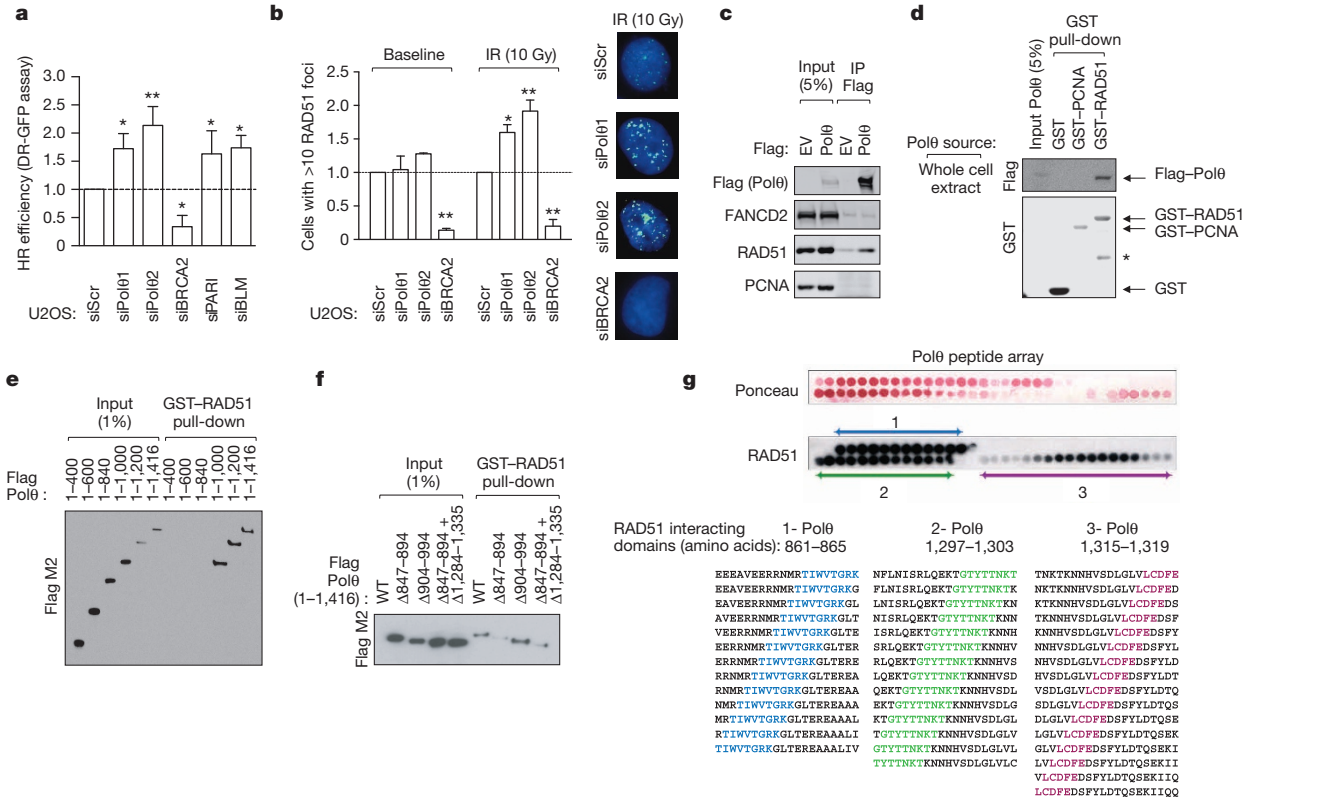


Figure 1 | Polθ is a RAD51-interacting protein that suppresses HR. **a**, DR-GFP assay in U2OS cells transfected with indicated siRNA. **b**, Quantification of RAD51 foci in U2OS cells transfected with indicated siRNA. **c**, Endogenous RAD51 co-precipitates *in vivo* with purified full-length Flag-tagged Polθ from whole cell extracts. EV, empty vector. **d**, GST pull-down experiment with full-length Flag-tagged Polθ (* indicates non-specific band). **e**, GST–RAD51 pull-down with *in vitro* translated Polθ truncation mutants. **f**, GST–RAD51 pull-down with *in vitro* translated Polθ versions missing indicated amino acids. **g**, Ponceau staining and immunoblotting of peptide arrays for the indicated Polθ motifs probed with recombinant RAD51. The Polθ amino acids spanning RAD51-interacting motifs are shown. Data in **a** and **b** represent mean ± s.e.m.

exposure, suggesting that Polθ regulates HR in cells under replicative stress (Extended Data Fig. 5a, b). Polθ-depleted cells were hypersensitive to cellular stress and DNA damage, along with an exacerbated checkpoint activation and increased γH2AX phosphorylation (Fig. 3b, c). Furthermore, the cell cycle progression of Polθ-depleted cells was impaired after DNA damage (Fig. 3d, e). To determine the role of Polθ in replication

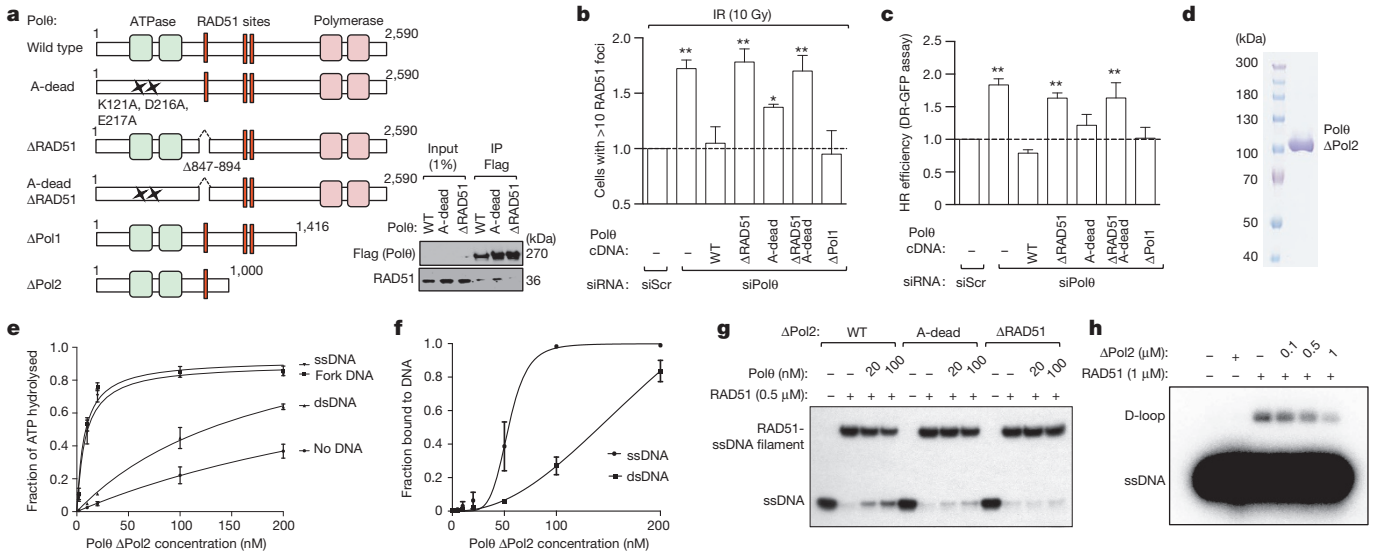


Figure 2 | Polθ inhibits RAD51-mediated recombination. **a**, Schematic of Polθ mutants used in complementation studies and their interaction with RAD51. WT, wild type. **b**, Quantification of RAD51 foci in U2OS cells transfected with indicated siRNA and Polθ cDNA constructs refractory to siPolθ1. **c**, DR-GFP assay in U2OS cells transfected with indicated siRNA and Polθ cDNA constructs refractory to siPolθ1. **d**, Coomassie-stained gel of the purified Polθ fragment. **e**, Quantification of Polθ ATPase activity. **f**, Quantification of Polθ binding to ssDNA and dsDNA. **g**, RAD51–ssDNA nucleofilament assembly assay. **h**, Assessment of RAD51-dependent D-loop formation. Data in **b**, **c**, **e** and **f** represent mean ± s.e.m.

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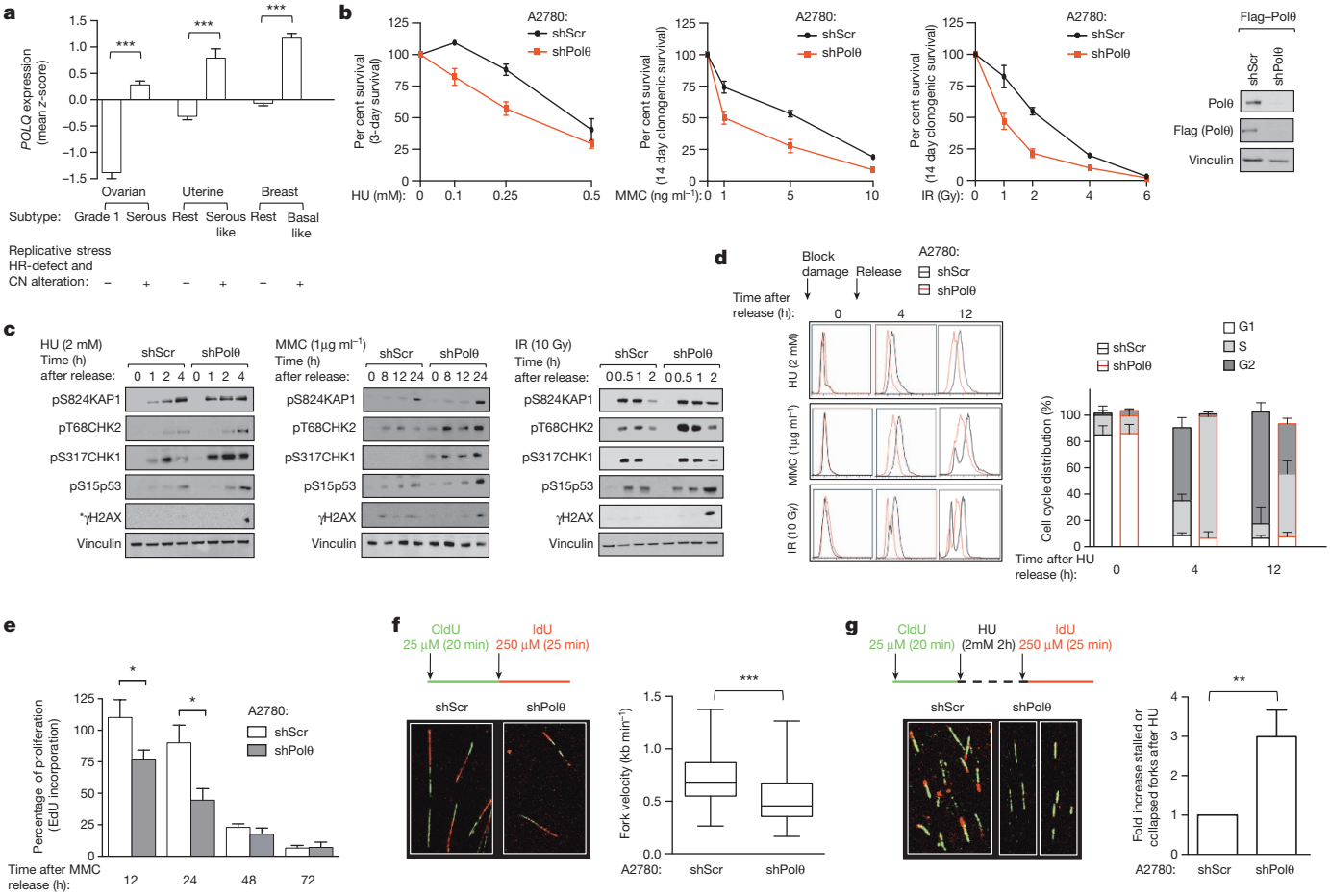


Figure 3 | Polθ promotes S phase progression and recovery of stalled forks. **a**, *POLQ* gene expression in subtypes of cancers with HR deficiency. **b**, Survival assays of A2780 cells exposed to the indicated DNA-damaging agents. Immunoblot showing silencing efficiency is shown on the right. **c**, Immunoblot analyses following pulse treatments with DNA-damaging agents (*γH2AX: see Methods for specific time points used for γH2AX immunoblot). HU, hydroxyurea. **d**, Cell cycle progression of synchronized A2780 cells. A representative cell cycle distribution. **e**, Fraction of cycling A2780 cells measured by 5-ethynyl-2′-deoxyuridine (EdU) incorporation. **f**, Quantification of DNA fibre lengths. **g**, Percentage of stalled forks. All experiments shown in **a**–**d** were performed in two cell lines (A2780 and 293T). All data represent mean ± s.e.m. except for box plots in **f** that show twenty-fifth to seventy-fifth percentiles, with lines indicating the median, and whiskers indicating the smallest and largest values.

dynamics, single-molecule analyses were performed on extended DNA fibres[19]. Abnormalities in replication fork progression were observed in Polθ-depleted cells (Fig. 3f, g and Extended Data Fig. 5c, d). These results suggest that Polθ maintains genomic stability at stalled or collapsed replication forks by promoting fork restart.

To examine the regulation of Polθ, we quantified Polθ expression by RT–qPCR. Polθ was selectively upregulated in HR-deficient ovarian cancer cell lines. Complementation of BRCA1 or FANCD2-deficient cell lines with BRCA1 and FANCD2 cDNA respectively, restored normal HR function and reduced Polθ expression to normal levels. Conversely, siRNA-mediated inhibition of HR genes increased Polθ expression (Extended Data Fig. 5e, f). Polθ expression was significantly higher in subgroups of cancers with HR deficiency and a high genomic instability pattern[20] (Fig. 3a and Extended Data Fig. 5g). Patients with high Polθ expression had a better response to platinum chemotherapy, a surrogate for HR deficiency, suggesting that Polθ expression inversely correlates with HR activity and may be useful as a biomarker for platinium sensitivity (Extended Data Fig. 5h, i). Together, these data indicate that increased Polθ expression is driven by HR deficiency.

To assess the possible synthetic lethality between HR genes and Polθ, we generated an HR-deficient ovarian tumour cell line, A2780-shFANCD2 cells (Extended Data Fig. 6a–c). These cells, and the parental A2780 cells, were subjected to Polθ depletion, and survival following exposure to cytotoxic drugs was measured. Polθ depletion reduced the survival of HR-deficient cells exposed to inhibitors of PARP (PARPi), cisplatin (CDDP) or MMC (Extended Data Fig. 6d–f). Polθ inhibition impaired the survival of BRCA1-deficient tumours (MDA-MB-436) after PARPi treatment but had no effect on the complemented line (MDA-MB-436 + BRCA1) (Fig. 4a). Polθ-depleted cells were hypersensitive to ATM inhibition, known to create an HR defect phenotype[21]. Chromosomal breakage, checkpoint activation, and γH2AX phosphorylation in response to MMC were exacerbated by Polθ depletion (Fig. 4b and Extended Data Fig. 6g, h). Furthermore, a whole-genome short hairpin RNA (shRNA) screen performed on HR-deficient (*FANCA* $^{-/-}$) fibroblasts showed that shRNAs targeting Polθ impair cell survival in MMC (Extended Data Fig. 6i), suggesting that HR-deficient cells cannot survive in the absence of Polθ.

Next, we investigated the interaction between the HR and Polθ pathways *in vivo* by interbreeding *Fancd2*$^{+/-}$ and *Polq*$^{+/-}$ mice. Although *Fancd2*$^{-/-}$ and *Polq*$^{-/-}$ mice are viable and exhibit subtle phenotypes[7,22], viable *Fancd2*$^{-/-}$*Polq*$^{-/-}$ mice were uncommon from these matings (Extended Data Fig. 7a). The only surviving *Fancd2*$^{-/-}$*Polq*$^{-/-}$ pups exhibited severe congenital malformations and were either found dead or died prematurely. *Fancd2*$^{-/-}$*Polq*$^{-/-}$ embryos showed severe congenital malformations, and mouse embryonic fibroblasts (MEFs) generated from *Fancd2*$^{-/-}$*Polq*$^{-/-}$ embryos showed hypersensitivity to PARPi (Fig. 4c and Extended Data Fig. 7b–e). These data suggest that loss of the HR and Polθ repair pathways *in vivo* results in embryonic lethality.

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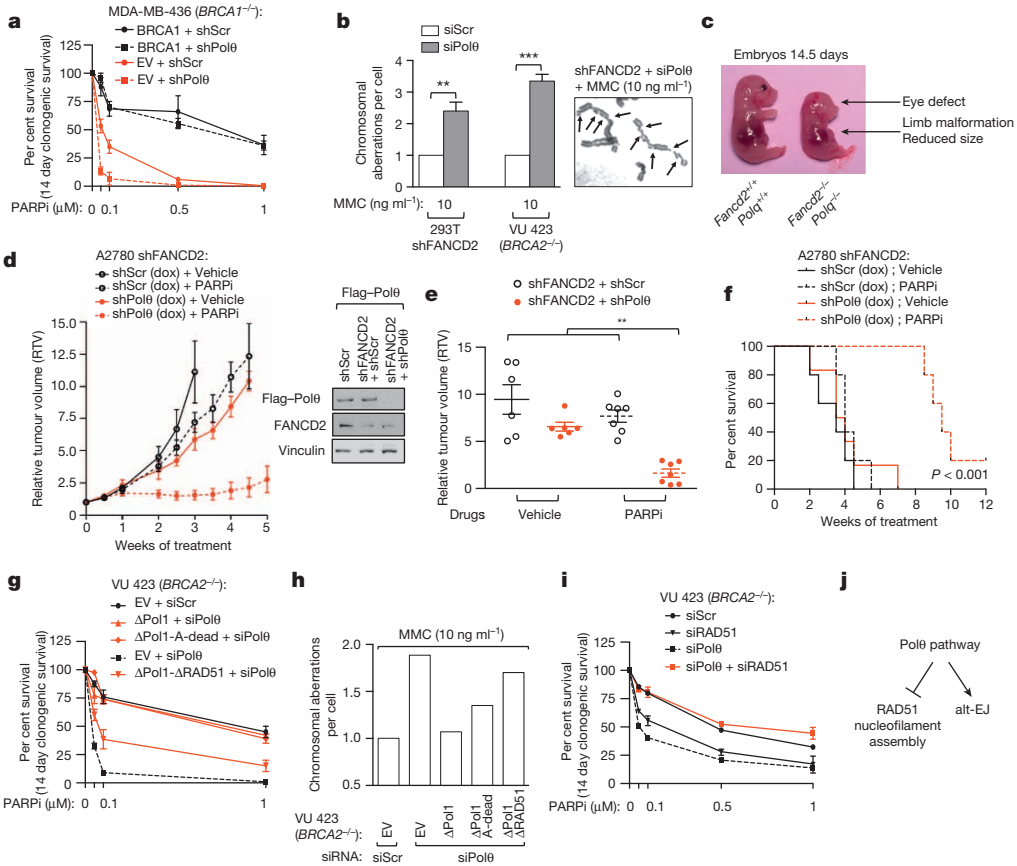


Figure 4 | Synthetic lethality between HR and Polθ repair pathways. **a**, Clonogenic formation of BRCA1-deficient (MDA-MB-436) cells expressing indicated cDNA together with indicated shRNA. **b**, Chromosome breakage analysis of HR-deficient cells transfected with the indicated siRNA. A representative image is shown. Arrows indicate chromosomal aberrations. **c**, Embryos at day 14 of gestation. **d**, Growth of indicated xenografts *in vivo*. Immunoblot showing silencing efficiency. **e**, Relative tumour volumes (RTV) for individual mice treated in **d** after three weeks of treatment. **f**, Overall survival for mice treated with vehicle or PARPi. Log-rank $P < 10^{-3}$. **g**, **h**, Clonogenic formation (**g**) and chromosome breakage analysis (**h**) of BRCA2-deficient cells expressing Polθ cDNA constructs refractory to siPolθ1 and transfected with the indicated siRNA. **i**, Clonogenic formation of BRCA2-deficient cells transfected with the indicated siRNA. **j**, Model for role of Polθ in DNA repair. Data in **a, b, g** and **i** represent mean ± s.e.m. For data in **d**–**f**, each circle represents data from one tumour and each group represents $n \geq 7$ tumours from $n \geq 6$ mice. Brackets show mean ± s.e.m.

As xenografts of tumours cells expressing shRNAs against both FANCD2 and Polθ did not stably propagate in mice (Extended Data Fig. 7f), we xenotransplanted A2780-shFANCD2 cells expressing either doxycycline-inducible Polθ or scrambled (Scr) shRNA in athymic nude mice. Polθ depletion significantly impaired tumour growth after PARPi treatment (Fig. 4d, e and Extended Data Fig. 7g, h). Moreover, mice bearing Polθ-depleted tumours had a survival advantage following PARPi treatment compared to control mice (Fig. 4f). Polθ-depleted HR-deficient tumour cells also exhibited decreased survival in *in vivo* dual-colour competition experiments (Extended Data Fig. 7i–l). Collectively, these data confirm that HR-deficient tumours are hypersensitive to inhibition of Polθ-mediated repair.

To understand which functions of Polθ are required for resistance to DNA-damaging agents, we performed a series of complementation studies in HR-deficient cells. Expression of full-length Polθ or ΔPol1, but not ΔRAD51, in HR-deficient Polθ-depleted cells treated with PARPi or MMC was able to rescue toxicity, suggesting that the anti-recombinase activity of Polθ maintains the genomic stability of HR-deficient cells (Fig. 4g, h and Extended Data Fig. 8a, b). Moreover, the toxicity induced by loss of Polθ in HR-deficient cells was rescued by depletion of RAD51 showing that, in the absence of Polθ, RAD51 is toxic to HR-deficient cells (Fig. 4i). These results suggest a role for Polθ in limiting toxic HR events[23] (Extended Data Fig. 8c–f) and may explain why HR-deficient cells overexpress and depend on an anti-recombinase for survival.

High mutation rates have been observed in HR-deficient tumours[24]. Previous studies have shown that Polθ is an error-prone polymerase[25,26] that participates in alternative end-joining (alt-EJ)[10]. Therefore, we assessed the role of Polθ in error-prone DNA repair in human cancer cells. Polθ inhibition reduced alt-EJ efficiency in U2OS cells, similar to the reduction observed following depletion of PARP1, another critical factor in end-joining[27,28] (Extended Data Fig. 9a). Expression of full-length Polθ, ΔRAD51, or A-dead, but not the ΔPol1 mutant, complemented the cells, suggesting that the polymerase domain of Polθ is required for end-joining (Extended Data Fig. 9b). GFP-tagged full-length Polθ formed foci after UV treatment in a PARP-dependent manner (Extended Data Fig. 9c). Polθ inhibition reduced the mutation frequency induced by UV light, and tumours with high Polθ expression harboured more somatic point mutations than those with lower Polθ levels (Extended Data Fig. 9d, e). These results suggest that Polθ contributes to the mutational signature observed in some HR-deficient tumours[29].

In human cancers, a deficiency in one DNA repair pathway can result in cellular hyper-dependence on a second compensatory DNA repair pathway[4]. Here we show that Polθ is overexpressed in EOCs and other tumours with HR defects[30]. Wild-type Polθ limits RAD51–ssDNA nucleofilament assembly (Extended Data Fig. 10a) and promotes alt-EJ (Fig. 4j). We demonstrate that HR-deficient tumours are hypersensitive to inhibition of Polθ-mediated repair. Therefore, Polθ appears to channel DNA repair by antagonizing HR and promoting PARP1-dependent error-prone repair (Extended Data Fig. 10b). These results offer a potential new therapeutic target for cancers with inactivated HR.

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Received 1 September; accepted 24 December 2014.
Published online 2 February 2015.

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Supplementary Information is available in the online version of the paper.

Acknowledgements We thank N. Shima, D. Chowdhury, G. Shapiro, J. Walter, D. Kozono, J. Ablain, M. Delord, J.-B. Lazaro and members of the D'Andrea laboratory for discussions. We also thank H. Kim for providing DNA constructs, L. Moldovan for providing GST tagged RAD51, S. Johnson for providing MDA-MB 436 cells, A. Y. Li for technical assistance, L. Moreau for chromosomal breakage analysis, and K. Mouw for critical reading of the manuscript. We thank H. Zhang for providing access to shRNA screening data shown in Extended Data Fig. 6i. We thank C. Cazaux and J.-S. Hoffmann for providing pcDNA Flag-tagged Polθ. R.C. received support from the Philippe Foundation and is a recipient of the Ovarian Cancer Research Fellowship (OCRF). This work was supported by NIH grants P50CA168504 and R01HL52725 and by grants from OCRF and BCRF.

Author Contributions R.C. conceived the study, performed experiments, and wrote the manuscript. J.C.L. and T.Y. purified Polθ fragments from insect cells and performed ATPase and gel shift assays. R.A. performed D-loop formation assays. I.H. and S.J.E. performed the DNA fibres assay. B.P. performed mice work and analysed *in vivo* data. M.I.R.P. and S.J.B. performed the Polθ peptide array and the RAD51–ssDNA filament assembly and release assays. K.W.O. scored RAD51 foci. P.A.K. curated TCGA datasets for Figure 3a and Extended Data Figures 5h and 9e and provided clinical perspectives. A.D.D. conceived the study and wrote the manuscript. All authors approved the final version of the manuscript.

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METHODS

Bioinformatic analysis. Gene Set Enrichment Analysis algorithm (GSEA, http://www.broadinstitute.org) was performed for the data sets summarized in Supplementary Table 1. TransLesion Synthesis (TLS) and polymerase gene sets are described in Supplementary Table 3. Row expression data were downloaded from Gene Expression Omnibus (GEO). Quantile normalizations were performed using the RMA routine through GenePattern. GSEA was run using GenePattern (http://www.broadinstitute.org) and corresponding *P* values were computed using 2,000 permutations. The DNA repair gene set used in Extended Data Fig. 1g has been determined according to a list of 151 DNA genes previously used[31]. GSEA analysis for 151 repair genes has been performed on the ovarian serous data sets (GSE14001, GSE14007, GSE18520, GSE16708, GSE10971). The list of 20 genes shown in Extended Data Fig. 1g represents the top 20 expressed gene in cancer samples (median of the 5 data sets). The waterfall plot in Extended Data Fig. 1h was generated as follows: the 20 genes defined in Extended Data Fig. 1g were used as a gene set; GSEA for indicated data sets was performed and the nominal *P* values were plotted. Supervised analysis of gene expression for GSE9891 was performed with respect to differential expression that differentiated the third of tumours with highest *POLQ* expression from the two-thirds with lowest *POLQ* levels. A list of the 200 most differentially expressed probe sets between the 2 groups (Supplementary Table 2) with false discovery rate <0.05 was analysed for biological pathways (hypergeometrical test; http://www.broadinstitute.org). TCGA data sets were accessed through the public TCGA data portal (https://tcga-data.nci.nih.gov/tcga/). Fig. 3a reflects *POLQ* gene expression in the ovarian carcinoma data set GSE9891, uterine carcinoma TCGA and breast carcinoma TCGA. Normalization of *POLQ* expression values across data sets was performed using *z*-score transformation. *POLQ* expression values were subdivided in subgroups reflecting the stage of the disease (for GSE9891: grade 3 ovarian serous carcinoma, $n = 143$ compared to type 1 (grade 1) ovarian cancers, $n = 20$; for uterine: serous-like tumours, $n = 60$ compared to the rest of the tumours, $n = 172$; for breast: basal-like breast carcinoma, $n = 80$ compared to the rest of the tumours, $n = 421$). Progression-free survival curves were generated by the Kaplan–Meier method and differences between survival curves were assessed for statistical significance with the log-rank test. In the absence of a clinically defined cut-off point for *POLQ* expression levels, we divided patients into 2 groups: those with *POLQ* mRNA levels equal to or above the median (*POLQ* high group) and those with values below the median (*POLQ* low group). We then analysed the correlation of *POLQ* with outcome in each group. Patients with cyclin E1 (CCNE1) amplification (resistant to CDDP) were excluded from the analysis. For mutation count, we accessed data from tumours included in the TCGA data sets for which gene expression and whole-exome DNA sequencing was available. Data were accessed through the public TCGA data portal and the cBioPortal for Cancer Genomics (http://www.cbioportal.org). For each TCGA data set, non-synonymous mutation count was assessed in tumours with the highest *POLQ* expression (top 33%) and compared to tumours with low *POLQ* expression (the remaining 67%). In the uterine TCGA[20], we curated all tumours except the ultra and hyper-mutated group (that is, POLE and MSI tumours). In the breast TCGA[32], all tumours were analysed. In the ovarian TCGA[1], we curated tumours harbouring molecular alterations (via mutation and epigenetic silencing) of the HR pathway.

Plasmid construction. To facilitate subcloning, a silent mutation (A390A) was introduced into the *POLQ* gene sequence to remove the unique Xho1 cutting site. Full-length or truncated *POLQ* cDNA were PCR-amplified and subcloned into pcDNA3-N-Flag, pFastBac-C-Flag, pOZ-C-Flag-HA, or GFP-C1 vectors to generate the various constructs. Point mutations and loop deletions were introduced by QuikChange II XL Site-Directed Mutagenesis Kit (Agilent Technologies) and confirmed by DNA sequencing. For Polθ rescue experiments (Fig. 4g, h and Extended Data Fig. 3d, e), *POLQ* cDNA constructs resistant to siPolθ1 were generated into the pOZ-C-Flag-HA vector and the constructs were stably expressed in indicated cell line by retroviral transduction. The Polθ ATPase catalytically-dead mutant (A-dead) was generated by mutating the Walker A and B motifs (Walker A: K121A and Walker B: D216A, E217A). pOZ-C-Flag-HA Polθ constructs were generated for retroviral transduction, and stable cells were selected using magnetic Dynabeads (Life Technologies) conjugated to the IL2R antibody (Millipore).

SiRNA and shRNA sequence information. For siRNA-mediated knockdown, the following target sequences were used: *POLQ* (Qiagen POLQ_1 used as siPolθ1 and Qiagen POLQ_6 used as siPolθ2); *BRCA1* (Qiagen BRCA1_13); *PARP1* (Qiagen PARP1_6); *REV1* (5′-CAGCGCAUCUGUGCCAAAGAA-TT-3′); *BRCA2* (5′-GAAGAAUGCAGGUUUAAUATT-3′); *BLM* (5′-AUCAGCUAGAGGCGAUCAATT-3′); *FANCD2* (5′-GGAGAUUGAUGGUCUACUATT-3′) and *PARI* (5′-AGGACACAUGUAAAGGGAUUGUCUATT-3′). AllStars negative control siRNA (Qiagen) served as the negative control. ShRNAs targeting human *FANCD2* was previously generated in the pTRIP/DU3-MND-GFP vector[33]. ShRNAs targeting human *POLQ* (CGGGCCTCTTTAGATATAAAT), human *BRCA2* (AAGAAGAATGCAGGTTTAATA) or control (Scr, scramble) were generated in the pLKO-1 vector. *POLQ* (V2THS_198349) and non-silencing TRIPZ-RFP doxycycline-inducible shRNA were purchased from Open Biosystems. All shRNAs were transduced using lentivirus.

Immunoblot analysis, fractionation and pull-down assays. Cells were lysed with 1% NP-40 lysis buffer (1% NP-40, 300 mM NaCl, 0.1 mM EDTA, 50 mM Tris (pH 7.5)) supplemented with protease inhibitor cocktail (Roche), resolved by NuPAGE (Invitrogen) gels, and transferred onto nitrocellulose membrane, followed by detection using the LAS-4000 Imaging system (GE Healthcare Life Sciences). For immunoprecipitation, cells were lysed with 300 mM NaCl lysis buffer, and the lysates were diluted to 150 mM NaCl before immunoprecipitation. Lysates were incubated with anti-Flag agarose resin (Sigma) followed by washes with 150 mM NaCl buffer. *In vitro* transcription and translation reactions were carried out using the TNT T7 Quick Coupled Transcription-Translation System (Promega). For cellular fractionation, cells were incubated with low-salt permeabilization buffer (10 mM Tris (pH 7.3), 10 mM KCl, 1.5 mM $MgCl_2$) with protease inhibitor on ice for 20 min. Following centrifugation, nuclei were resuspended in 0.2 M HCl and the soluble fraction was neutralized with 1 M Tris-HCl (pH 8.0). Nuclei were lysed in 150 mM NaCl and following centrifugation, the chromatin pellet was digested by micrococcal nuclease (Roche) for 5 min at room temperature. Recombinant GST–RAD51 and GST–PCNA fusion protein were expressed in BL21 strain and purified using glutathione-Sepharose beads (GE Healthcare) as previously described[15]. Beads with equal amount of GST or GST–RAD51 were incubated with *in vitro* translated Flag-tagged Polθ variants in 150 mM NaCl lysis buffer.

Antibodies and chemicals. Antibodies used in this study included: anti-PCNA (PC-10), anti-FANCD2 (FI-17), anti-RAD51 (H-92), anti-GST (B14), and histone H3 (FL-136) and anti-vinculin (H-10) (Santa Cruz); anti-Flag (M2) (Sigma); anti-pS317CHK1 (2344), anti-pT68CHK2 (2661) (Cell signalling); anti-pS824KAP-1 (A300-767A) (Bethyl); anti-pS317γH2AX (05636) (Millipore); anti-pS15p53 (ab1431) and anti-Polθ (ab80906) (abcam); anti-BrdU (555627) (BD Pharmingen). Mitomycin C (MMC), cis-diamminedichloroplatinum(II) (cisplatin, CDDP), and hydroxyurea (HU) were purchased from Sigma. The PARPi rucaparib (AG-014699) was purchased from Selleckchem and ABT-888 from AbbVie. Rucaparib was used for all *in vitro* assays and ABT-888 was used for all *in vivo* experiments.

Chromosomal breakage analysis. 293T and VU 423 cells were twice-transfected with siRNAs for 48 h and incubated for 48 h with or without the indicated concentrations of MMC. For complementation studies on 293T shFANCD2, *POLQ* cDNA constructs were transfected 24 h after the first siRNA transfection. Cells were exposed for 2 h to 100 ng ml^{-1} of colcemid and treated with a hypotonic solution (0.075 M KCl) for 20 min and fixed with 3:1 methanol/acetic acid. Slides were stained with Wright's stain and 50 metaphase spreads were scored for aberrations. The relative number of chromosomal breaks was calculated relative to control cells (si Scr). Radial figures were excluded from the analysis for clarity in Fig. 4b.

Reporter assays and immunofluorescence. HR and alt-EJ efficiency was measured using the DR-GFP (HR efficiency) and the alt-EJ reporter assay, performed as previously described[14,27,34]. Briefly, 48 h before transfection of SceI cDNA, U20S-DR-GFP cells were transfected with the indicated siRNA or PARPi (1 μM). The HR activity was determined by FACS quantification of viable GFP-positive cells 96 h after SceI was transfected. For RAD51 immunofluorescence experiments, cells were transfected with indicated siRNA 48 h before treatment with HU (2 mM) or IR (10 Gy). For complementation studies, Polθ cDNA constructs were either transfected 24 h after siRNA transfection (Fig. 2b, c and Extended Data Fig. 9b) or stably expressed in the indicated cell line (Extended Data Fig. 3d, e). 6 h after HU or IR treatment, cells were fixed with 4% paraformaldehyde for 10 min at room temperature, followed by extraction with 0.3% Triton X-100 for 10 min on ice. Antibody staining was performed at room temperature for 1 h. For quantification of RAD51 foci in BrdU positive cells, cells were transfected with indicated siRNA 48 h before treatment with IR (10 Gy). Then 2 h after IR treatment, cells were treated with BrdU pulse (10 μM) for 2 h and subsequently fixed with 4% paraformaldehyde and stained for RAD51 as described above. Cells were then fixed in ethanol (4 °C, overnight), treated with 1.5 M HCl for 30 min and stained for BrdU antibody. The relative number of cells with more than 10 RAD51 foci was calculated relative to control cells (si Scr). Statistical differences between cells transfected with siRNAs (si Polθ1, si Polθ2, si BRCA2, si PARI or si BLM relative to control (si Scr) were assessed. For GFP fluorescence, cells were grown on coverslips, treated with UV (24 h after GFP–Polθ transfection; 20 J m^{-2}), fixed with 4% paraformaldehyde for 10 min at 25 °C 4 h after the UV treatment, washed three times with PBS and mounted with DAPI-containing mounting medium (Vector Laboratories). When indicated, cells were treated with PARPi (1 μM) 24 h before GFP–Polθ transfection. Images were captured using a Zeiss AX10 fluorescence microscope and AxioVision software. Cells with GFP foci were quantified by counting number of cells with more than five foci. At least 150 cells were counted for each sample.

Cell survival assays. For assessing cellular cytotoxicity, cells were seeded into 96-well plates at a density of 1,000 cells per well. Cytotoxic drugs were serially diluted

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in media and added to the wells. At 72 h, CellTiter-Glo reagent (Promega) was added to the wells and the plates were scanned using a luminescence microplate reader. Survival at each drug concentration was plotted as a percentage of the survival in drug-free media. Each data point on the graph represents the average of three measurements, and the error bars represent the standard deviation. For clonogenic survival, 1,000 cells per well were seeded into 6-well plates and treated with cytotoxic drugs the next day. For MMC and PARPi, cells were treated continuously with indicated drug concentrations. For CDDP, cells were treated for 24 h and cultured for 14 days in drug-free media. Colony formation was scored 14 days after treatment using 0.5% (w/v) crystal violet in methanol. Survival curves were expressed as a percentage ± s.e.m. over three independent experiments of colonies formed relative to the DMSO-treated control.

Cell cycle analysis. A2780 cells expressing Scr or Polθ shRNA were synchronized by a double thymidine block (Sigma) and subsequently exposed to MMC (1 μg ml^{-1} for 2 h), IR (10 Gy) or HU (2 mM, overnight). At the indicated time points following drug release, cells were fixed in chilled 70% ethanol, stored overnight at −20 °C, washed with PBS, and resuspended in propidium iodide. A fraction of those cells was analysed by immunoblotting for DNA damage response proteins. The immunoblot analysis of γH2AX shows staining after 0, 24, 48 and 72 h of HU treatment. For proliferation experiments, cells were incubated with 5-ethynyl-2′-deoxyuridine (EdU) (10 μM) for 1 h at each time point after MMC exposure (1 μg ml^{-1} for 2 h). Cells were washed and resuspended in culture medium for 2 h before be analysed by flow cytometry. Edu staining was performed using the Click-iT EdU kit (Life Technologies).

DNA fibre analysis. A2780 cells expressing Scr or Polθ shRNA were incubated with 25 μM chlorodeoxyuridine (CldU) (Sigma, C6891) for 20 min. Cells were then treated with 2 mM hydroxyurea (HU) for 2 h and incubated in 250 μM iododeoxyuridine (IdU) (Sigma, I7125) for 25 min after washout of the drug. Spreading of DNA fibres on glass slides was done as previously reported[19]. Glass slides were then washed in distilled water and in 2.5 M HCl for 80 min followed by three washes in PBS. The slides were incubated for 1 h in blocking buffer (PBS with 1% BSA and 0.1% NP-40) and then for 2 h in rat anti-BrdU antibody (1:250, Abcam, ab6326). After washing with blocking buffer, the slides were incubated for 2 h in goat anti-rat Alexa 488 antibody (1:1,000, Life Technologies, A-11006). The slides were then washed with PBS and 0.1% NP-40 and then incubated for 2 h with mouse anti-BrdU antibody diluted in blocking buffer (1:100, BD Biosciences, 347580). Following an additional wash with PBS and 0.1% NP-40, the fibres were stained for 2 h with chicken anti-mouse Alexa 594 (1:1000, Life Technologies, A-21201). At least 150 fibres were counted per condition. Pictures were taken with an Olympus confocal microscope and the fibres were analysed by ImageJ software. The number of stalled or collapsed forks were measured by DNA fibres that had incorporated only CIdU. Stalled or collapsed forks counted in Polθ-depleted cells is expressed as fold-change after HU treatment relative to the fold-change observed in control cells, which was arbitrarily set to 1.

SupF mutagenesis assay. 293T cells twice-transfected with siRNAs for 48 h were then transfected with undamaged or damaged (UVC, 1,000 J m^{-2}) pSP189 plasmids using GeneJuice (Novagen). After 48 h, plasmid DNA was isolated with a miniprep kit (Promega) and digested with DpnI. After ethanol precipitation, extracted plasmids were transformed into the β-galactosidase− MBM7070 indicator strain through electroporation (GenePulsor X Cell; Bio-Rad) and plated onto LB plates containing 1 mM IPTG, 100 μg ml^{-1} 5-bromo-4-chloro-3-indolyl-β-D-galactopyranoside and 100 μg ml^{-1} ampicillin. White and blue colonies were scored using ImageJ software, and the mutation frequency was calculated as the ratio of white (mutant) to total (white plus blue) colonies.

***POLQ* gene expression.** RNA samples extracted using the TRIzol reagent (Invitrogen) were reverse transcribed using the Transcriptor Reverse Transcriptaze kit (Roche) and oligo dT primers. The resulting cDNA was use to analysed *POLQ* expression by RT-qPCR using with QuantiTect SYBRGreen (Qiagen), in an iCycler machine (Bio-Rad). *POLQ* gene expression values were normalized to expression of the housekeeping gene *GAPDH*, using the ΔCt method and are shown on a $\log_2$ scale. The primers used for *POLQ* are as follows: *POLQ* primer 1 (forward: 5′-TATCTG CTGGAACTTTGCTGA-3′; reverse: 5′-CTCACACCATTTCTTTGATGGA-3′); *POLQ* primer 2 (forward: 5′-CTACAAGTGAAGGGAGATGAGG-3′; reverse: 5′-TCAGAGGGTTTCACCAATCC-3′).

Polθ purification from insect SF9 cells. A Polθ fragment (ΔPol2) containing the ATPase domain with a RAD51 binding site (amino acids 1 to 1,000) was cloned into pFastBac-C-Flag and purified from baculovirus-infected SF9 insect cells as previously described[35]. Briefly, SF9 cells were seeded in 15-cm dishes at 80–90% confluency and infected with baculovirus. Three days post-infection, cells were collected and lysed in 500 mM NaCl lysis buffer (500 mM NaCl, 0.01% NP-40, 0.2 mM EDTA, 20% glycerol, 1 mM DTT, 0.2 mM PMSF, 20 mM Tris (pH 7.6)) supplemented with Halt protease inhibitor cocktail (Thermo Scientific) and calpain I inhibitor (Roche) and the protein was eluted in lysis buffer supplemented with 0.2 mg ml^{-1} of Flag peptide (Sigma). The protein was concentrated in lysis buffer using 10 kDa centrifugal filters (Amicon). The protein was quantified by comparing its staining intensity (Coomassie-R250) with that of BSA standards in a 7% Tris-glycine SDS–PAGE gel. Purified protein was flash-frozen in small aliquots in liquid nitrogen and stored at −80 °C.

Radiometric ATPase assay. Each 10 μl reaction consisted of 200 nM ATP, reaction buffer (20 mM Tris-HCl (pH 7.6), 5 mM $MgCl_2$, 0.05 mg ml^{-1} BSA, 1 mM DTT), and 5 μCi of [γ-^{32}P]ATP. For corresponding reactions, ssDNA, dsDNA, and forked DNA were added to the reaction in excess at a final concentration of 600 nM. Once all of the non-enzymatic reagents were combined, recombinant Polθ was added to start the ATPase reaction. After incubation for 90 min at room temperature, stop buffer (125 mM EDTA (pH 8.0)) was added and approximately ~0.05 μCi was spotted onto PEI-coated thin-layer chromatography (TLC) plates (Sigma). Unhydrolyzed [γ-^{32}P]ATP was separated from the released inorganic phosphate [$^{32}P_i$] with 1 M acetic acid, 0.25 M lithium chloride as the mobile phase. TLC plates were exposed to a phosphor screen and imaged with the BioRad Imager PMC. ssDNA, dsDNA, and forked DNA were generated as previously described[35]. To remove any contaminating ssDNA, dsDNA and forked DNA were gel purified after annealing. Spots corresponding to [γ-^{32}P]ATP and the released inorganic phosphate [$^{32}P_i$] were quantified (in units of pixel intensity) and the fraction of ATP hydrolysed calculated for each Polθ concentration.

Electrophoretic mobility gel shift assay (EMSA). Binding of Polθ to ssDNA was assessed using EMSA. 60-mer single-stranded DNA (ssDNA) or double-stranded DNA (dsDNA) oligonucleotides (5 nM) were incubated with increasing amount of Polθ (0, 5, 10, 50, or 100 nM) in 10 μl of binding buffer (20 mM HEPES-K$^+$, (pH 7.6), 5 mM magnesium acetate, 0.1 μg μl^{-1} BSA, 5% glycerol, 1 mM DTT, 0.2 mM EDTA, and 0.01% NP-40) for 1 h on ice. Polθ protein was added at a tenfold dilution so that the final salt concentration was approximately 50 mM NaCl. The ssDNA probes were 5′ fluorescently labelled with IRDye-700 (IDT). After incubation, the samples were analysed on a 5% native polyacrylamide/0.5X TBE gel at 4 °C. A fluorescent imager (Li-Cor) was used to visualize the samples in the gel.

RAD51 purification. Human GST–RAD51 was purified from bacteria as described[36]. *Xenopus* Rad51 (xRad51) was purified as follows. N-terminally His-tagged SUMO–Rad51 was expressed in BL21 pLysS cells. Three hours after induction with 1 mM IPTG, cells were collected and resuspended in buffer A (50 mM Tris-Cl (pH 7.5), 350 mM NaCl, 25% sucrose, 5 mM β-mercaptoethanol, 1 mM PMSF and 10 mM imidazole). Cells were lysed by supplementation with Triton X-100 (0.2% final concentration), three freeze–thaw cycles and sonication (20 pulses at 40% efficiency). The soluble fraction was separated by centrifugation and incubated with 2 ml of Ni-NTA resin (Qiagen) for 1 h at 4 °C. After washing the resin with 100 ml of wash buffer (buffer A supplemented with 1 M NaCl, final concentration), the salt concentration was brought down to 350 mM. His–SUMO–Rad51 was eluted with a linear gradient of imidazole from 10–300 mM in buffer A. Eluted fractions were analysed by SDS–PAGE. His–SUMO–Rad51 containing fractions were pooled and supplemented with Ulp1 protease to cleave the His–SUMO tag and dialysed overnight into buffer B (50 mM Tris-Cl (pH 7.5), 350 mM NaCl, 25% sucrose, 10% glycerol, 5 mM β-mercaptoethanol, 10 mM imidazole and 0.05% Triton X-100). The dialysed fraction was incubated with Ni-NTA resin for 1 h at 4 °C and the Rad51 containing flow-through fraction was collected and dialysed overnight into buffer C (100 mM potassium phosphate (pH 6.8), 150 mM NaCl, 10% glycerol, 0.5 mM DTT and 0.01% Triton-X). Rad51 was further purified by hydroxyapatite (Bio-Rad) chromatography. After washing with ten column volumes of buffer C, Rad51 was eluted with a linear gradient of potassium phosphate (pH 6.8) from 100–800 mM. Rad51 containing fractions were analysed by SDS–PAGE and dialysed into storage buffer (20 mM HEPES-KOH (pH 7.4), 150 mM NaCl, 10% glycerol, 0.5 mM DTT). Purified protein was flash-frozen in small aliquots in liquid nitrogen and stored at −80 °C.

D-loop assay. D-loop formation assays were performed using xRad51 and conducted as previously described[37]. Briefly, nucleofilaments were first formed by incubating RAD51 (1 μM) with end-labelled 90-mer ssDNA (3 μM nt) at 37 °C for 10 min in reaction buffer containing 20 mM HEPES-KOH (pH 7.4), 1 mM ATP, 1 mM $MgCl_2$, 1 mM DTT, BSA (100 μg ml^{-1}), 20 mM phosphocreatine and creatine phosphokinase (20 μg ml^{-1}). After the 10 min incubation, increasing amounts of Polθ (0, 0.1, 0.5, or 1.0 μM) and RPA (200nM) were added and incubated for an additional 15 min at 37 °C. The reaction was then supplemented with 1 mM $CaCl_2$ followed by further incubation at 37 °C for 15 min. D-loop formation was initiated by the addition of supercoiled dsDNA (pBS-KS (−), 79 μM bp) and incubation at 37 °C for 15 min. D-loops were analysed by electrophoresis on a 0.9% agarose gel after deproteinization. Gel was dried and exposed to a PhosphoImager (GE Healthcare) screen for quantification.

Substitution peptide arrays and RAD51-ssDNA filament experiments. Substitution peptide arrays were performed as previously described[17]. RAD51 displacement assays were performed as follows. Binding reactions (10 μl) contained 5′-^{32}P-end-labelled

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DNA substrates (0.5 ng of 60 mer ssDNA) and various amounts of human RAD51 and/or Polθ in binding buffer (40 mM Tris-HCl (pH 7.5), 50 mM NaCl, 10 mM KCl, 2 mM DTT, 5 mM ATP, 5 mM MgCl2, 1 mM DTT, 100 mg ml^{-1} BSA) were conducted at room temperature. After 5 min incubation with Polθ and a further 5 min incubation with RAD51 or vice versa, an equimolar amount of cold DNA substrate was added to the reaction. Products were then analysed by electrophoresis through 10% PAGE (200 V for 40 min in 0.5× Tris-borate-EDTA buffer) and visualized by autoradiography.

Interbreeding of the *Fancd2* and *Polq* mice. For the characterization of *Fancd2*/*Polq* conditional knockouts, we crossed C57BL/6J mice (Jackson Laboratory). $Fancd2^{+/-}Polq^{+/+}$ mice, previously generated in our laboratory[22], were crossed with $Fancd2^{+/+}Polq^{+/-}$ mice[7] to generate $Fancd2^{+/-}Polq^{+/-}$ mice. These double heterozygous mice were then interbred, and the offspring from these mating pairs were genotyped using PCR primers for *Fancd2* and *Polq*. A statistical comparison of the observed with the predicted genotypes was performed using a two-sided Fisher's exact test. Primary MEFs were generated from E13.5 to E15 embryos and cultured in RPMI supplemented with 15% fetal bovine serum and 1% penicillin-streptomycin. All data generated in the study were extracted from experiments performed on primary MEFs from passage 1 to passage 4. The primers used for mice genotyping are as follows: *Fancd2* PCR primers OST2cF (5′-CATGCATATAGGAACCCGAAGG-3′), OST2aR (5′-CAGGACCTTTGGAGAAGCAG-3′) and LTR2bF (5′-GGCGTTACTTAAGCTAGCTTG-3′); *Polq* PCR primers IMR5973 (5′-TGCAGTGTACAGATGTTACTTTT-3′), IMR 5974 (5′-TGGAGGTAGCATTTCTTCTC-3′), IMR 5975 (5′-TCACTAGGTTGGGGTTCTC-3′) and IMR 5976 (5′-CATCAGAAGCTGACTCTAGAG-3′). Specific PCR conditions are available upon request.

Studies of xenograft-bearing CrTac:NCr-Foxn1nu mice. The Animal Resource Facility at The Dana-Farber Cancer Institute approved all housing situations, treatments and experiments using mice. No more than five mice were housed per air-filtered cage with *ad libitum* access to standard diet and water, and were maintained in a temperature- and light-controlled animal facility under pathogen-free conditions. All mice described in this text were drug and procedure naive before the start of the experiments. For every xenograft study, we subcutaneously implanted approximately 1.0×10^6 A2780 cells (1:1 in Matrigel Matrix, BD Biosciences) into both flanks of 6–8-week-old female CrTac:NCr-Foxn1nu mice (Taconic). Doxycycline (Sigma) was added to the food (625 p.p.m.) and bi-weekly (Tuesday and Friday) to the water (200 μg ml^{-1}) for mice bearing tumours that reached 100–200 mm^3. Roughly one week (5–6 days) after the addition of doxycycline to the diet, mice were randomized to twice daily treatment schedules with vehicle (0.9% NaCl) or PARPi (ABT-888; 50 mg per kg body weight) by oral gavage administration for the indicated number of weeks. Overall survival was determined using Kaplan–Meier analyses performed with log-rank tests to assess differences in median survival for each shRNA condition (shScr or shPolθ) and each treatment condition (vehicle or PARPi) (GraphPad Prism 6 Software). For competition assays, A2780 cells expressing FANCD2-GFP shRNA (GFP positive cells) or a combination of FANCD2–GFP shRNA with (doxycycline inducible) Scr-RFP or Polθ-RFP shRNA (GFP–RFP positive cells) were mixed at an equal ratio of GFP to GFP–RFP positive cells, and thereafter injected into nude mice given doxycycline-containing diets and treated with either vehicle or PARPi or CDDP. For competition assays, mice received identical doxycycline and PARPi drug treatment. For the cisplatin competition assay, mice were randomized into semi-weekly treatment regimens with vehicle (0.9% NaCl) or CDDP (5 mg per kg body weight) by intraperitoneal injection. After three to four weeks of treatment, mice were euthanized and tumours were grown *in vitro*, in the presence of doxycycline (2 μg ml^{-1} for 4 days). The relative ratio of GFP to GFP–RFP positive cells was determined by FACS analysis. Tumour volumes were calculated bi-weekly using caliper measurements (length × $width^2$)/2. Growth curves were plotted as the mean tumour volume (mm^3) for each treatment group; relative tumour volume (RTV) indicates the change in tumour volume at a given time point relative to the tumour volume at the day of initial measurement (volume of approximately 0.15 cm^3) which was arbitrarily set to 1. Mice were unbiasedly assigned into different treatment groups. Drug treatment and outcome assessment was performed in a blinded manner. Mice were monitored every day and euthanized by CO_2 inhalation when tumour size (≥ 2 cm), tumour status (necrosis/ulceration) or body weight loss (≥ 20%) reached ethical endpoint, according to the rules of the Animal Resource Facility at The Dana-Farber Cancer Institute.

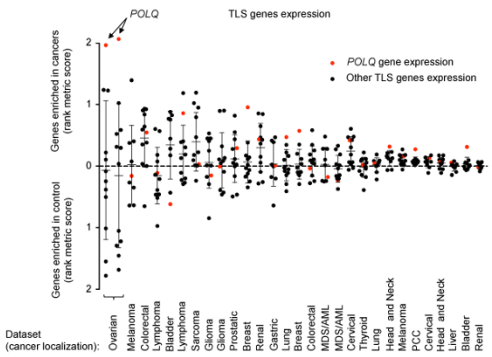
Immunohistochemical staining. We stained formalin-fixed paraffin-embedded sections of harvested xenografts with antibodies specific for γH2AX (pSer139) (Upstate Biotechnology) and Ki67 (Dako). At least two xenografts were scored for each treatment. Tumours were collected 3 weeks after treatment. At least five 40 × fields were scored. The mean ± s.e.m. percentage of positive cells from five images in each treatment group was calculated.

Statistical analysis. Unless stated otherwise, all data are represented as mean ± s.e.m. over at least three independent experiments, and significance was calculated using the Student's *t*-test. Asterisks indicate statistically significant ($*P < 0.05$; $**P < 10^{-2}$; $***P < 10^{-3}$) values. All the *in vivo* experiments were run with at least 6 tumours from 6 mice for each condition. No statistical methods were used to predetermine sample size.

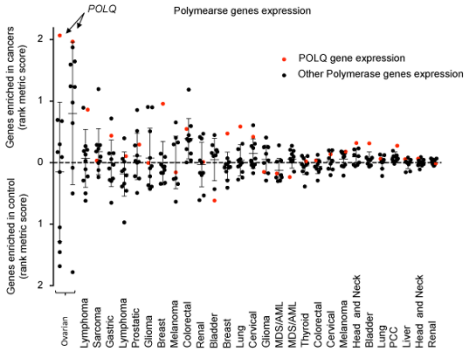
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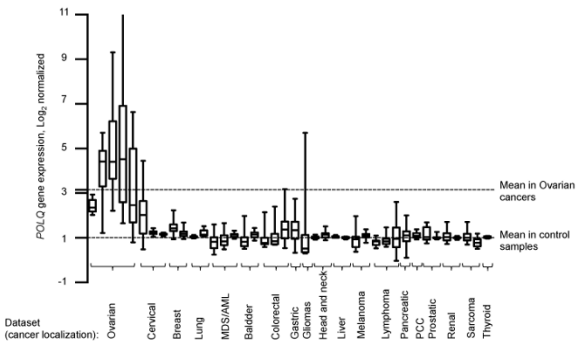
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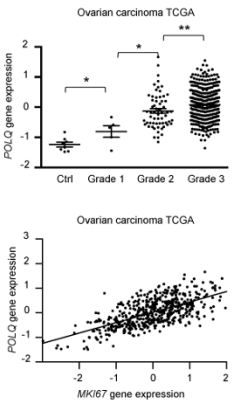
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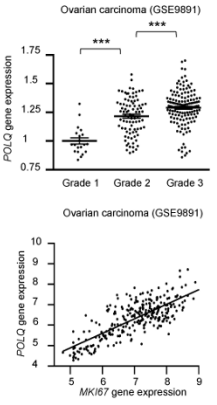
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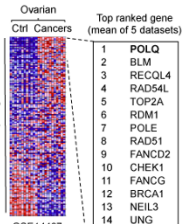
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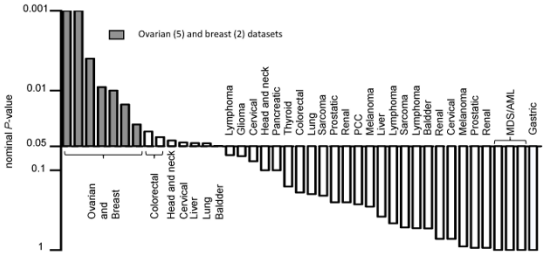
Gene Ontology (GO) Biological process	Number of GO term	*P* value (mean)
Cell cycle	8	1,51942E-05
Mitosis and cytokinesis	46	5,44857E-05
Homologous recombination	4	5,27823E-05
Checkpoint	3	6,00749E-07

g

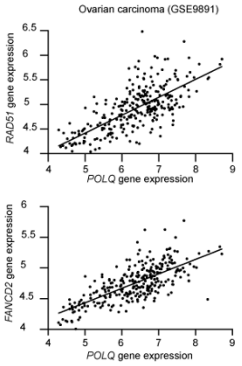


Top ranked gene (mean of 5 datasets)	
1	**POLQ**
2	BLM
3	RECQL4
4	RAD54L
5	TOP2A
6	RDM1
7	POLE
8	RAD51
9	FANCD2
10	CHEK1
11	FANCG
12	BRCA1
13	NEIL3
14	UNG
15	FANCA
16	RAD54B
17	APEX2
18	PRKDC
19	FANCB
20	FANCE

h



i



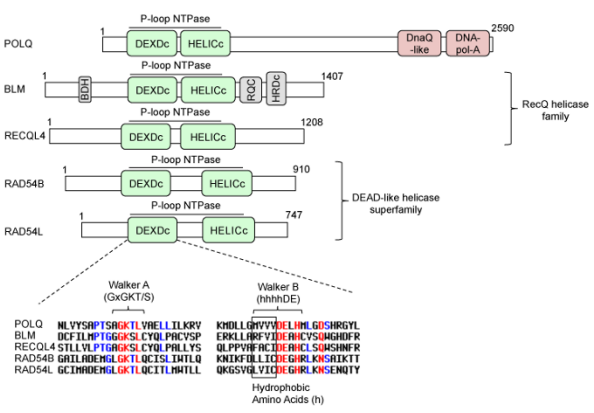
j

POLQ	DNA repair activity
TOPBP1	yes
BLM	yes
RAD54L	yes
PLK4	no
FANCI	yes
SMC4	no
BRCA1	yes
FANCD2	yes
ATAD5	yes
TRAIP	no

k

Gene Ontology (GO) molecular function	GO ID	*P* value
DNA helicase activity	GO:0003678	0.00002
helicase activity	GO:0004386	0.0001
nucleoside-triphosphatase activity	GO:0017111	0.0002
pyrophosphatase activity	GO:0016462	0.0002
hydrolase activity, acting on acid anhydrides	GO:0016817	0.0002
hydrolase activity, acting on acid anhydrides, in phosphorus-containing anhydrides	GO:0016818	0.0002
catalytic activity	GO:0003824	0.0003
DNA-directed DNA polymerase activity	GO:0003887	0.0003
DNA polymerase activity	GO:0034061	0.0005
damaged DNA binding	GO:0003684	0.0009

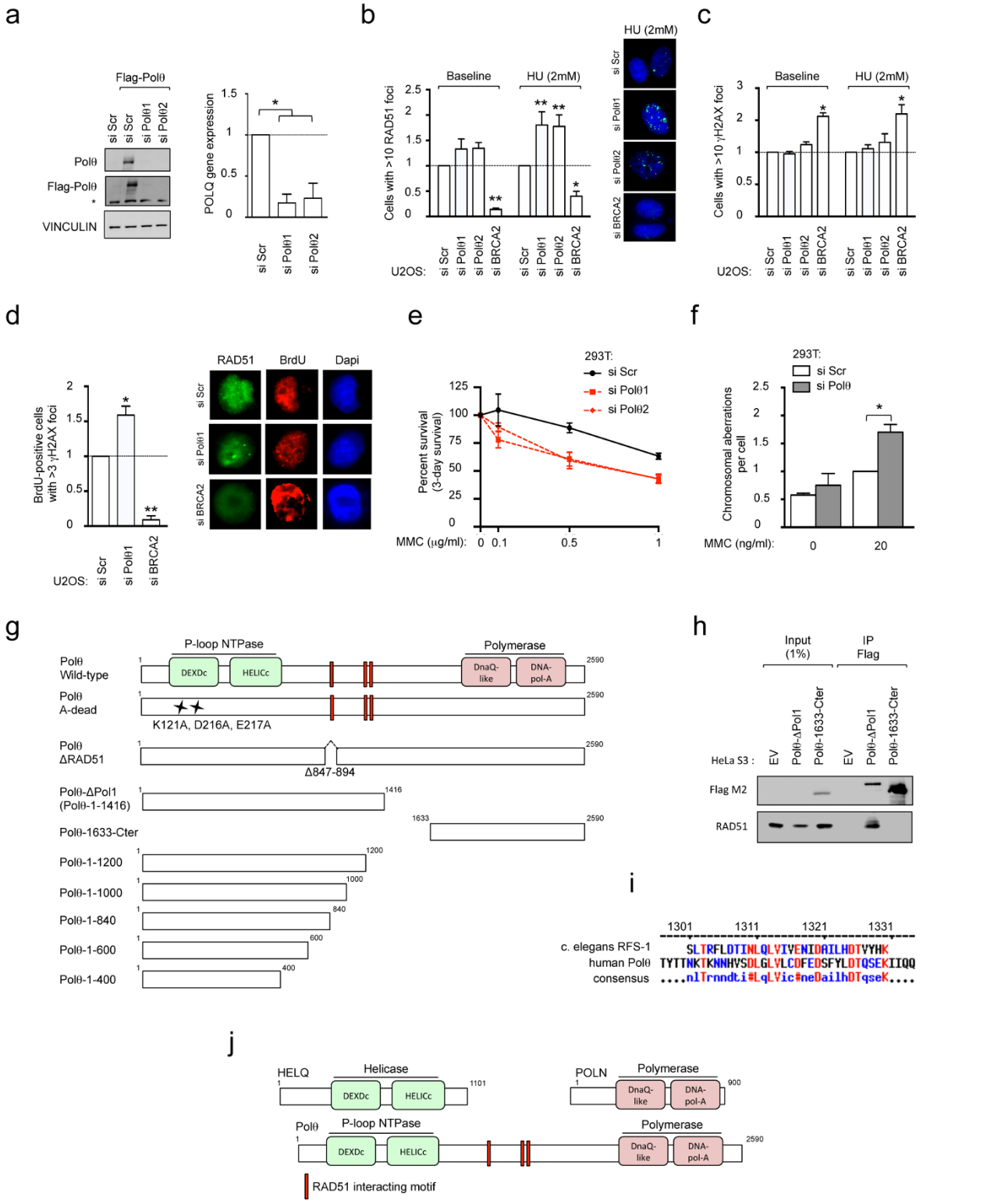
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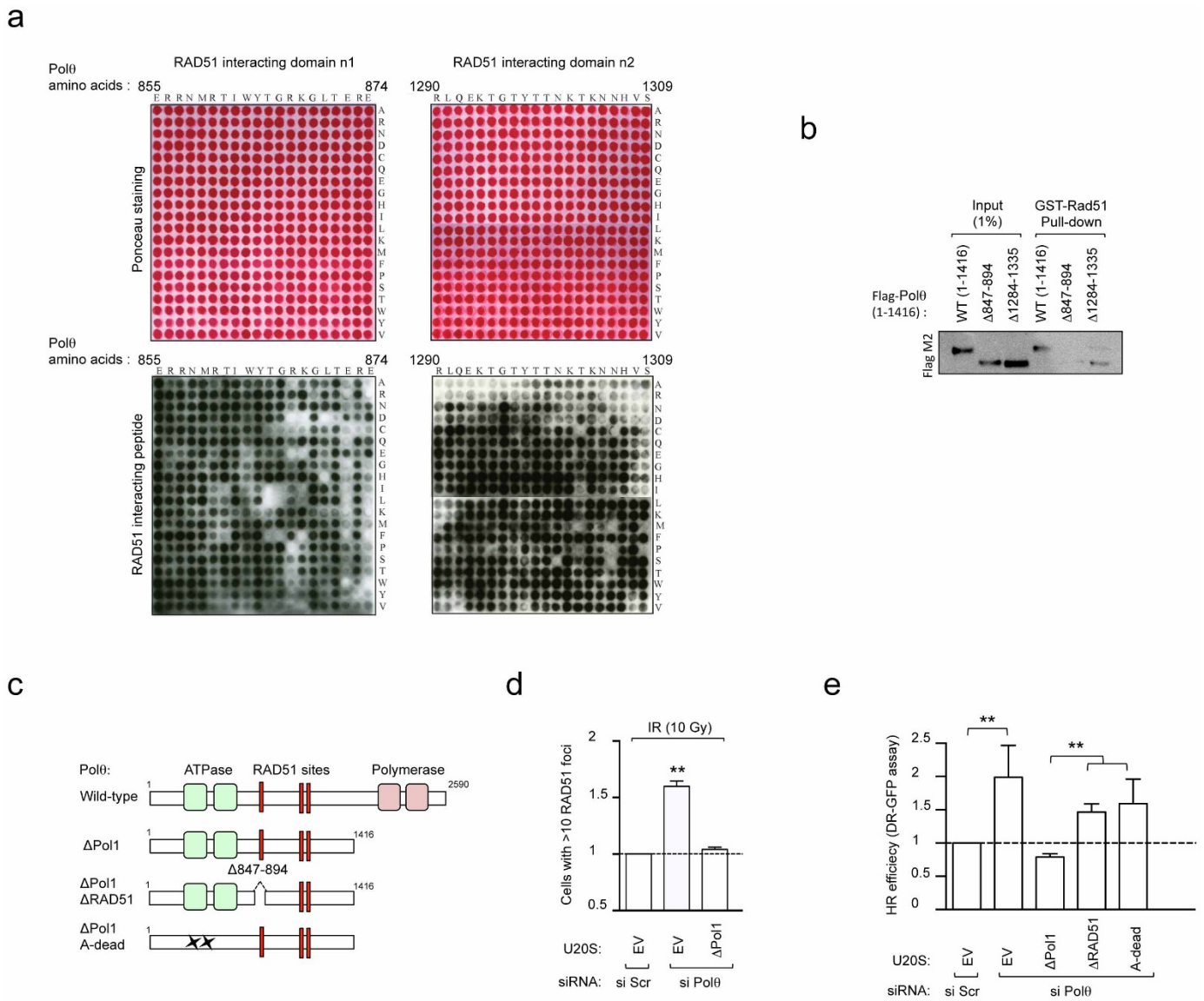
Extended Data Figure 1 | *POLQ* is highly expressed in epithelial ovarian cancers (EOCs) and *POLQ* expression correlates with expression of HR genes. **a**, **b**, Gene set enrichment analysis (GSEA) for expression of TransLesion Synthesis (TLS) (**a**) and polymerase (**b**) genes between primary cancers and control samples in 28 independent data sets from 19 different cancers types. Enrichment values (represented as a single dot for each gene in a defined data set) were determined using the rank metric score to compare expression values between cancers and control samples. Dots above the dashed line reflect enrichment in cancer samples, whereas dots below the dashed line show gene expression enriched in control samples. Data sets were ranked based on the amplitude of the rank metric score and plotted as shown. **c**, *POLQ* gene expression in 40 independent data sets from 19 different cancer types. For each data set, *POLQ* values were expressed as fold-change differences relative to the mean expression in control samples, which was arbitrarily set to 1. **d**, *POLQ* expression correlates with tumour grade and MKi67 gene expression in the ovarian TCGA ($n = 494$ patients with ovarian carcinoma (grade 1, $n = 5$; grade 2, $n = 61$; grade 3, $n = 428$) and control samples, $n = 8$). **e**, *POLQ* expression correlates with tumour grade MKi67 gene expression in the ovarian data set GSE9891 ($n = 251$ patients with ovarian serous and endometrious carcinoma for which grade status was available (grade 1, $n = 20$; grade 2, $n = 88$; grade 3, $n = 143$)). Statistical correlation was assessed using the Pearson test (for **d**: $r = 0.65$, $P < 10^{-3}$; for **e**: r = 0.77, $P < 10^{-3}$). **f**, Top-ranked biological pathways differentially expressed between samples expressing high levels of *POLQ* (high *POLQ*, first 33%, $n = 95$) relative to samples with low *POLQ* expression (low *POLQ*, 67%, $n = 190$) on the ovarian data set GSE9891 ($n = 285$ patients with ovarian carcinoma). Significance values were determined by the hypergeometrical test using the 200 most differentially expressed probe sets between the 2 groups (high *POLQ* and low *POLQ*). **g**, GSEA for expression of DNA repair genes between primary cancers and control samples in 5 independent ovarian cancer data sets. A representative heat map showing differential gene expression between ovarian cancers and controls is shown from GSE14407. For each data set, DNA repair genes were ranked based on the metric score reflecting their enrichment in cancer samples. The top 20 DNA repair genes primarily expressed in cancer samples compared to control samples is shown on the right. **h**, GSEA for the top 20 DNA repair genes defined in **g** between primary cancers and control samples in 40 independent cancer data sets. The nominal *P* value was used as a measure of the expression enrichment in cancer samples and represented as a waterfall plot. When the gene set expression was enriched in control samples, the *P* value was arbitrarily set to 1. **i**, *POLQ* expression correlates with *RAD51* and *FANCD2* gene expression in 285 samples from the ovarian data set GSE9891. Statistical correlation was assessed using the Pearson test ($r = 0.71$, $P < 10^{-3}$). **j**, Top 10 genes that most closely correlated with *POLQ* expression (gene neighbours analysis) for 1,046 cell lines from the CCLE collection. DNA repair activity for these genes is indicated in the table. Increased HR gene expression is known to positively correlate with improved response to platinum based chemotherapy (a surrogate of HR deficiency) and thus can be predictive of decreased HR activity[31,38]. Conceptually, a state of HR deficiency may lead to compensatory increased expression of other HR genes. **k**, Top-ranked Gene Ontology (GO) terms for the molecular functions encoded by the top 20 DNA repair genes defined in Extended Data Fig. 1g. **l**, Schematic representation of Polθ domain structure with the helicases (BLM, RECQL4, RAD54B and RAD54L) that co-expressed with Polθ (from Extended Data Fig. 1g). Conserved amino-acid sequences of ATP binding and hydrolysis motifs (namely Walker A and B) are indicated. Cox plots in **c** show twenty-fifth to seventy-fifth percentiles, with lines indicating the median, and whiskers indicating the smallest and largest values. For **d** and **e** (top panels), each dot represents the expression value from one patient, brackets show mean ± s.e.m.

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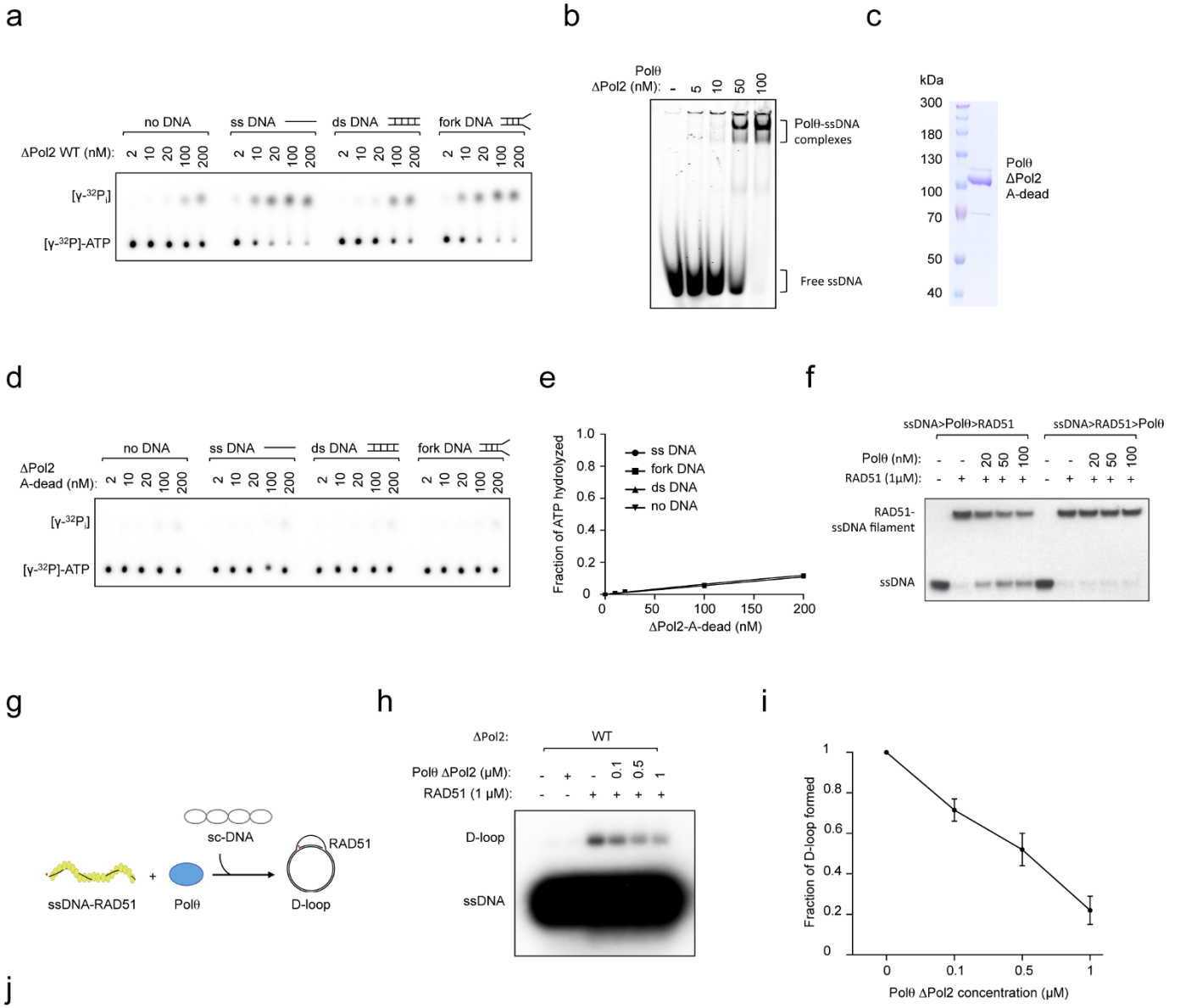
Extended Data Figure 2 | Polθ is a RAD51-interacting protein required for maintenance of genomic stability. **a**, siRNA sequences (siPolθ1 and siPolθ2) efficiently downregulate exogenously transfected Polθ protein. Polθ levels were detected by immunoblotting with Flag or Polθ antibody (left) and by RT–qPCR using 2 different sets of *POLQ* primers (right). The asterisk on the immunoblot indicates a non-specific band. Expression was normalized using *GAPDH* as a reference gene. *POLQ* gene expression values are displayed as fold-change differences relative to the mean expression in control cells, which was arbitrarily set to 1. **b**, Quantification of baseline and HU-induced RAD51 foci in U2OS cells transfected with the indicated siRNA. **c**, Quantification of baseline and HU-induced γH2AX foci in U2OS cells transfected with the indicated siRNA. **d**, Quantification of IR-induced RAD51 foci in BrdU-positive U2OS cells transfected with the indicated siRNA. **e**, Polθ inhibition by siRNA induced a decrease in the cellular survival of 293T cells treated with MMC in a 3-day survival assay. **f**, Quantification of chromosomal aberrations in 293T cells transfected with the indicated siRNA. **g**, Schematic representation of Polθ truncation proteins used for RAD51 interaction studies. **h**, Endogenous RAD51 co-precipitates with Flag-tagged Polθ-ΔPol1 (Polθ-1–1,416) but not Polθ-1633-Cter, each stably expressed in HeLa cells. **i**, Sequence alignment between the RAD51-interacting motifs of *C. elegans* RFS-1 and human Polθ. **j**, Schematic of Polθ domain structure with its homologues HELQ and POLN. All data show mean ± s.e.m.

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Extended Data Figure 3 | Characterization of RAD51-interacting motifs in Polθ. **a**, Substitution peptide array probed with recombinant RAD51 and analysed by immunoblotting. A 20-mer peptide spanning each of the RAD51 binding sites (shown in Fig. 1g) were created in which each amino acid of the original peptide was mutated to each of the 20 amino acids and RAD51 binding activity was tested. The amino acid change for each of the amino acids of the RAD51 interacting domain of Polθ is shown on the right. Ponceau staining was used to visualize position of the peptides within the array. **b**, GST–RAD51 pull-down with *in vitro* translated Polθ proteins missing indicated amino acids. **c**, Schematic of Polθ mutants used in complementation studies. **d**, Quantification of IR-induced RAD51 foci in U2OS cells stably integrated with empty vector (EV) or Polθ-ΔPol1 cDNA that is refractory to siPolθ1. Cells were transfected with indicated siRNA and subsequently treated with IR. The number of cells with more than 10 RAD51 foci was calculated relative to control cells (si Scr). **e**, DR-GFP assay in U2OS cells stably integrated with empty vector (EV) or indicated Polθ cDNA constructs refractory to siPolθ1 and transfected with indicated siRNA. All data show mean ± s.e.m.

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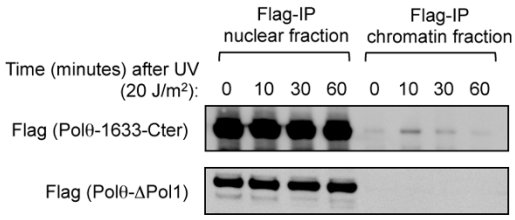


Conditions tested	RAD51 foci	DR-GFP	RAD51-ssDNA assembly	D-loop formation
si Polθ	↑	↑	NA	NA
Polθ cDNA WT	↓	↓	↓	↓
Polθ-A-dead cDNA	-	-	-	NA
Polθ-ΔRAD51 cDNA	-	-	-	NA

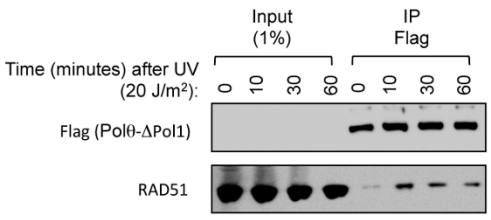
Extended Data Figure 4 | Polθ is an ATPase that suppresses RAD51–ssDNA nucleofilament assembly and formation of RAD51-dependent D-loop structures. a, Representative ΔPol2 wild-type radiometric ATPase assay. **b**, Gel mobility shift assays with ΔPol2 wild type and ssDNA. **c**, Coomassie-stained gel showing the purified ΔPol2-A-dead fragment. **d**, Representative ΔPol2-A-dead radiometric ATPase assay. **e**, Quantification of ΔPol2-A-dead ATPase activity. (ssDNA, single-stranded DNA; dsDNA, double-stranded DNA). **f**, Assembly/disruption of RAD51-ssDNA filaments in the presence of increasing amounts of ΔPol2 wild type. The order in which each component was added to the reaction is noted above. **g**, Schematics of the formation of RAD51-dependent D-loop structures. **h** Formation of RAD51-containing D-loop structures following the addition of increasing amounts of ΔPol2 wild type. **i**, Fraction of D-loop formed following the addition of increasing amounts of ΔPol2 wild type. **j**, Effect of siPolθ and the different Polθ cDNA constructs on HR read-out. NA, not applicable. Data in **i** shows mean ± s.e.m.

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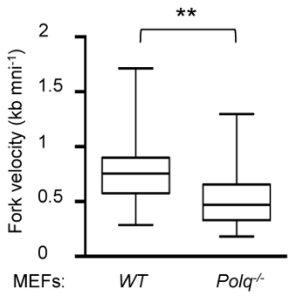
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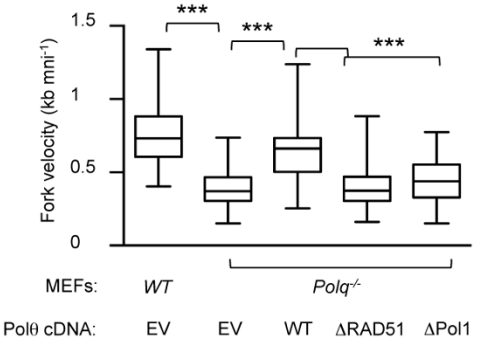
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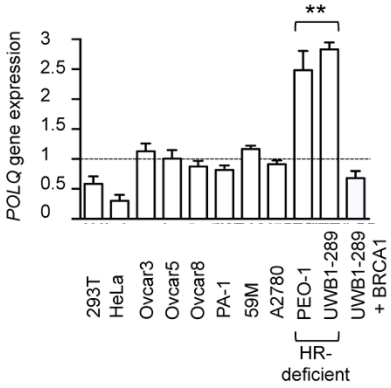
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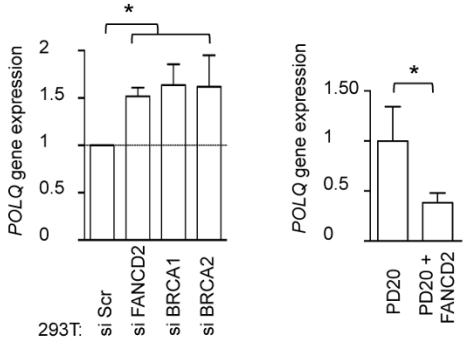
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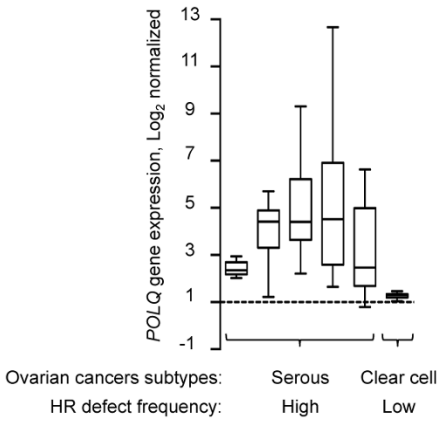
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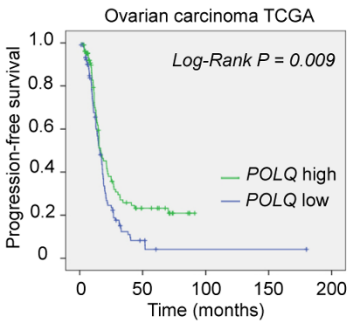
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g



h



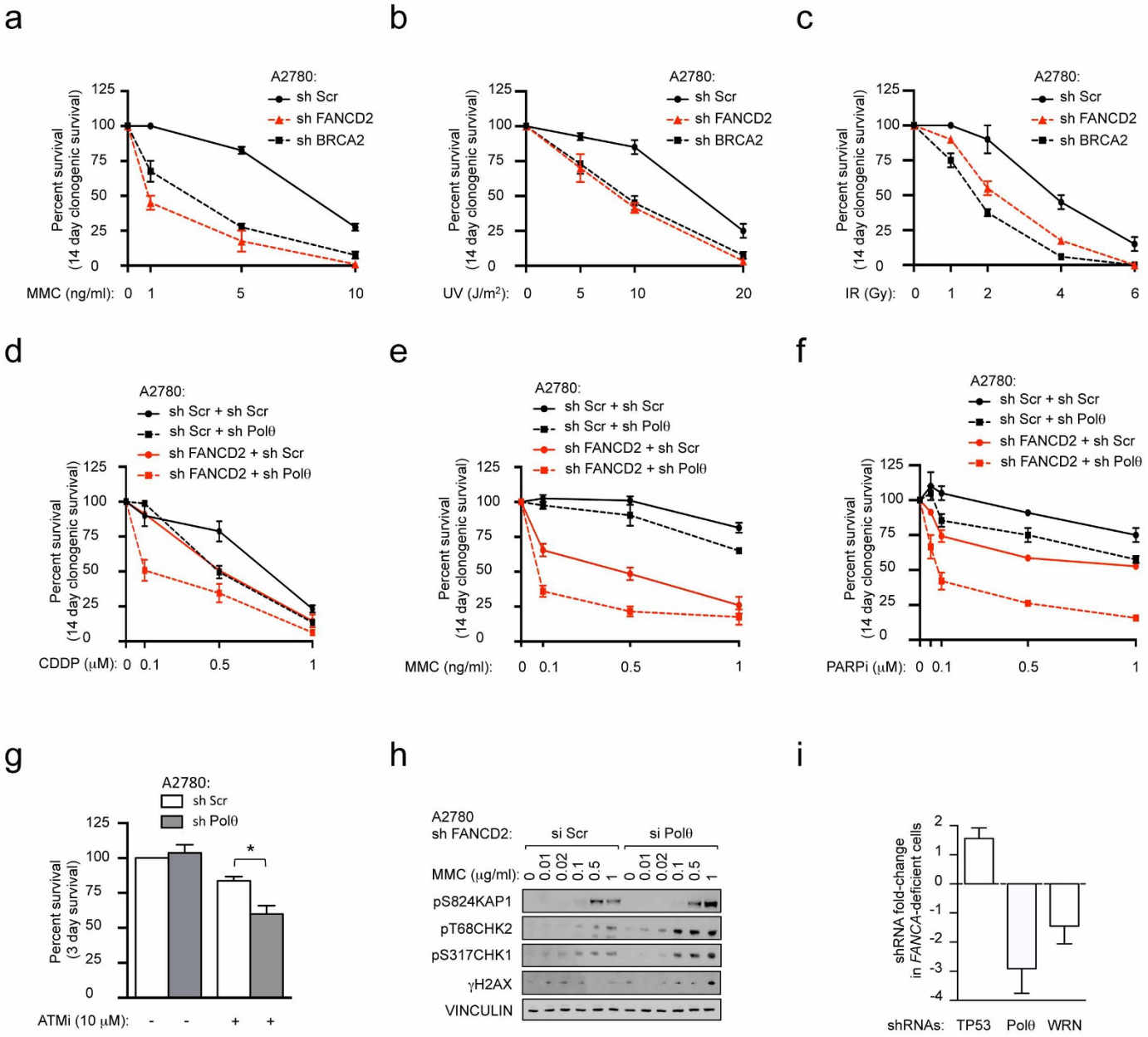
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	POLQ expression	HR activity	Tumors sensitivity to CDDP or PARPi
Validated experimentaly	-	+	-
	+	-	++
	-	-	++++
Hypothetical based	+	NA	++

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Extended Data Figure 5 | Polθ functions under replicative stress and is induced by HR deficiency. **a**, Polθ recruitment to the chromatin is enhanced by UV treatment. HeLa cells stably integrated with either Flag-tagged ΔPol1 or Polθ-1633-Cter (Extended Data Fig. 2g) were subjected to UV treatment. Cells were collected at indicated time points after UV treatment and IPs were performed on nuclear and chromatin fractions. **b**, HeLa cells stably integrated with ΔPol1 were treated with UV and collected at the indicated time points following UV exposure. Polθ and RAD51 co-precipitation is enhanced by UV treatment. **c**, Quantification of DNA fibre lengths isolated from wild-type or *Polq*$^{-/-}$ MEFs. **d**, Quantification of DNA fibre lengths isolated from wild-type or *Polq*$^{-/-}$ MEFs transfected with either EV or Polθ cDNA constructs. **e**, *POLQ* gene expression was analysed by RT–qPCR in HR-deficient ovarian cancer cell lines (PEO-1 and UWB1-289) compared with other ovarian cancer cell lines, HeLa (cervical cancer) cells and 293T (transformed human embryonic kidney) cells. Expression was normalized using *GAPDH* gene as a reference. *POLQ* expression values are displayed as fold-change relative to the mean expression in HR-proficient control cells, which was arbitrarily set to 1. **f**, *POLQ* gene expression analysis (RT–qPCR) in 293T cells transfected with siRNA targeting *FANCD2*, *BRCA1* or *BRCA2* (left panel) and in corrected PD20 cells (PD20 + FANCD2) relative to FANCD2-deficient cells (PD20) (right panel). Expression was normalized using *GAPDH* gene as a reference. *POLQ* expression values are presented as fold-change relative to the mean expression in control cells, which was arbitrarily set to 1. **g**, *POLQ* gene expression in 5 data sets of serous epithelial ovarian carcinoma (frequently associated with an HR deficiency) and 1 data set of clear cell ovarian carcinoma (subgroup not associated with HR alterations). For each data set, *POLQ* expression values are displayed as fold-change differences relative to the mean expression in control samples, which was arbitrarily set to 1. **h**, Progression-free survival (PFS) after first line platinum chemotherapy for patients with ovarian carcinoma (ovarian carcinoma TCGA). Statistical significance was assessed by the log-rank test ($P < 10^{-2}$). **i**, Effect of Polθ expression levels and HR status on tumour senstivity to cisplatin or PARPi. NA, not applicable. Box plots in **c**, **d**, and **g** show twenty-fifth to seventy-fifth percentiles, with lines indicating the median, and whiskers indicating the smallest and largest values. Data in **e** and **f** show mean ± s.e.m.

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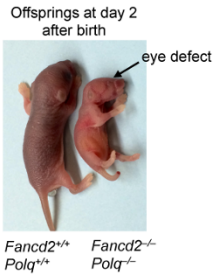
Extended Data Figure 6 | Polθ inhibition sensitizes HR-deficient tumours to cytotoxic drug exposure. **a**–**c**, Clonogenic formation of A2780 cells expressing scrambled (Scr) shRNA or shRNAs against FANCD2 or BRCA2 with increasing amounts of MMC (**a**), UV (**b**) or IR (**c**). **d**–**f**, Clonogenic formation of A2780 cells expressing scrambled (Scr) or FANCD2 shRNA, together with shRNA targeting Polθ, in increasing concentrations of CDDP (**d**), MMC (**e**) or PARPi (**f**). **g**, Inhibition of Polθ reduces the survival of A2780 cells after 3 days of continuous exposure to the ATM inhibitor Ku55933. **h**, Immunoblot analyses in A2780 cells expressing FANCD2 shRNA together with siRNA targeting Polθ or Scr at 24 h after indicated MMC pulse treatment. **i**, FANCA-deficient fibroblasts (GM6418) were infected with a whole-genome shRNA library and treated with MMC for 7 days. The fold-change enrichment of each shRNA after MMC treatment was determined by sequencing relative to the infected cells before treatment. TP53 depletion is known to improve survival of *FANCA*$^{-/-}$ cells[33]. WRN depletion has recently been shown to be synthetically lethal with HR deficiency[39]. Each column represents the mean of at least 2 independent shRNAs. All data show mean ± s.e.m.

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a

Polq status	*Fancd2* status	Offspring observed (n)	% observed	% expected	Significant difference
+/+	+/+	19	7.2	6.25	no
+/+	+/-	43	16.2	12.5	no
+/+	-/-	22	8.3	6.25	no
+/-	+/+	36	13.6	12.5	no
+/-	+/-	61	23.0	25	no
+/-	-/-	25	9.4	12.5	no
-/-	+/+	21	7.9	6.25	no
-/-	+/-	34	12.8	12.5	no
-/-	-/-	0^{ψ}	0	6.25	yes, $P < 0.001$
Total number :		261	100	100	

b



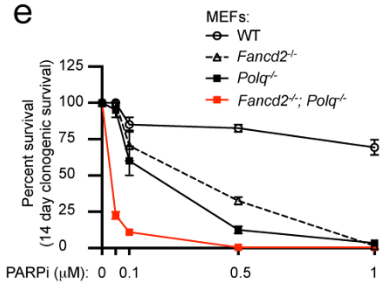
c

Polq status	*Fancd2* status	Embryos observed (n)	% observed	% expected	Significant difference
+/+	+/+	13	6.6	6.25	no
+/+	+/-	31	15.7	12.5	no
+/+	-/-	7	3.6	6.25	no
+/-	+/+	22	11.2	12.5	no
+/-	+/-	62	31.5	25	no
+/-	-/-	28	14.2	12.5	no
-/-	+/+	10	5.1	6.25	no
-/-	+/-	16	8.1	12.5	no
-/-	-/-	8	4.1	6.25	no
Total number :		197	100	100	

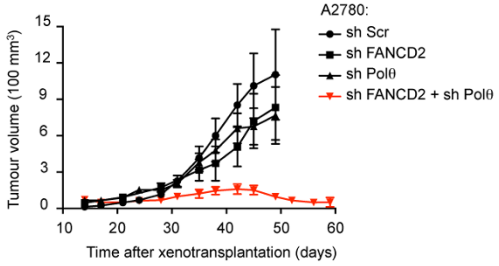
d

Malformation observed	% of *Polq*$^{-/-}$ *Fancd2*$^{-/-}$ embryos observed with malformations
Reduced body weight	100
Reduced body size	100
Eye defect	100
Limb malformation	12.5

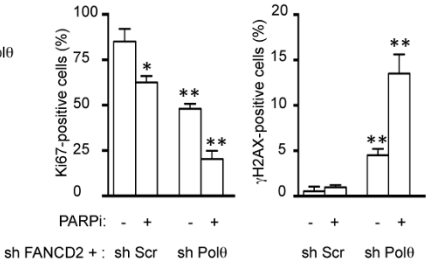
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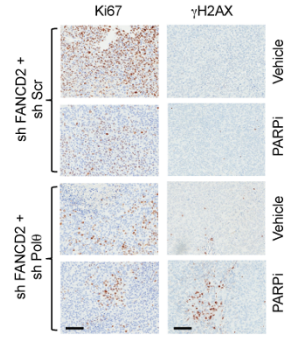
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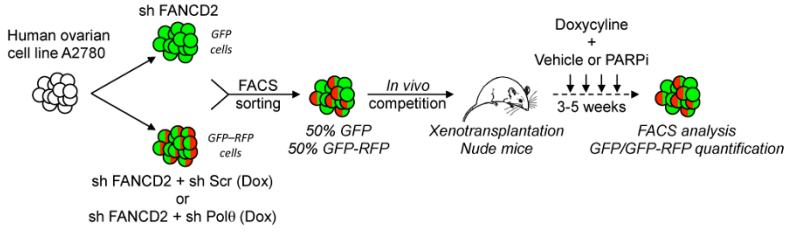
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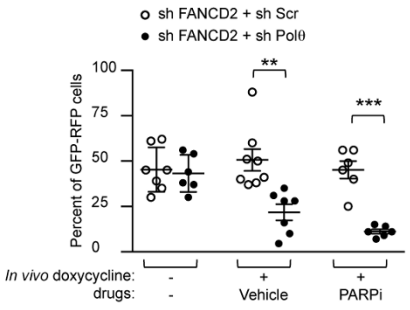
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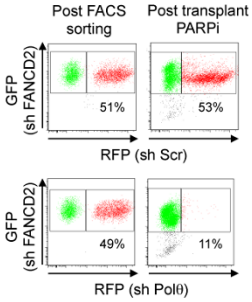
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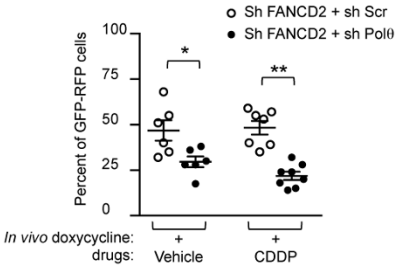
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k



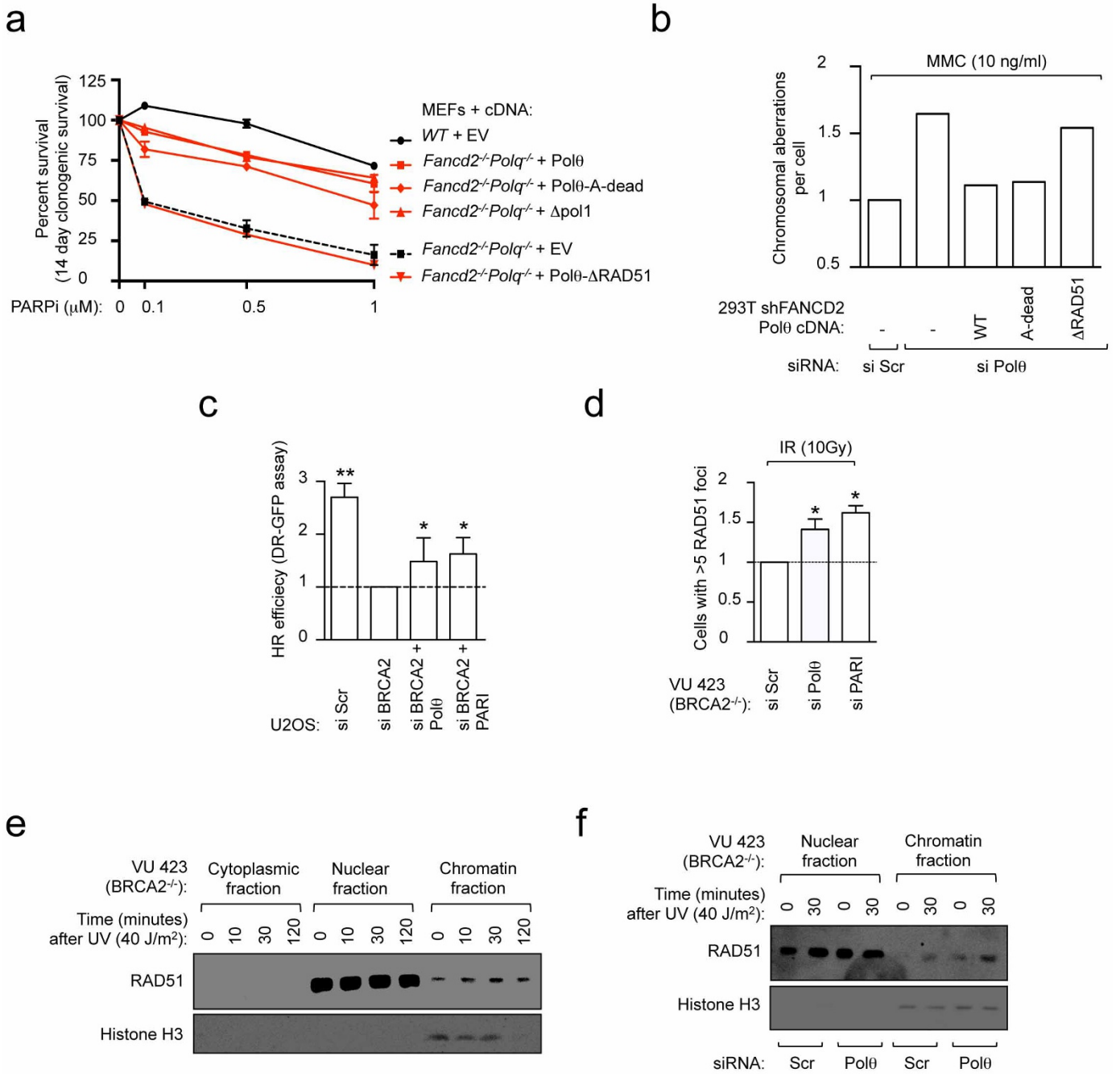
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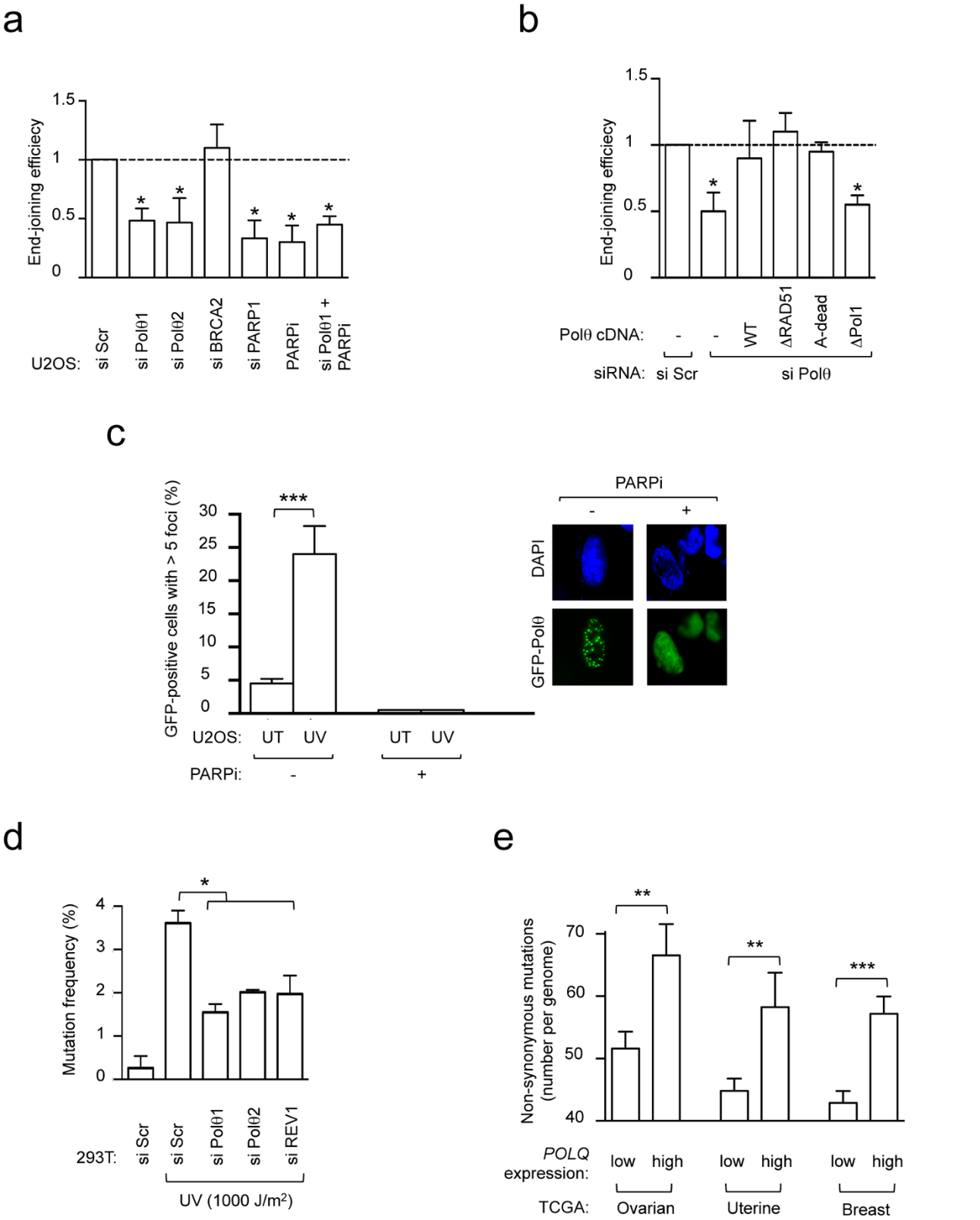
Extended Data Figure 7 | HR and Polθ repair pathways are synthetical lethal *in vivo*. **a**, Genotypes frequencies of offspring from interbred *Fancd2*$^{+/-}$*Polq*$^{+/-}$ mice. Ψ, four *Fancd2*$^{-/-}$*Polq*$^{-/-}$ offsprings were observed with several congenital malformations and premature death within 48 h of birth. **b**, Description of *Fancd2*$^{-/-}$*Polq*$^{-/-}$ offspring generated in the study. The offspring presented congenital malformations (that is, eye defects) together with reduced size and body weight. The arrow indicates absence of the right eye. **c**, Genotypes frequencies of E13.5 to E15 embryos (13.5 to 15 days post coitum) from interbred *Fancd2*$^{+/-}$*Polq*$^{+/-}$ mice. **d**, Description of congenital malformations and their measured frequencies observed in E13.5 to E15 *Fancd2*$^{-/-}$*Polq*$^{-/-}$ embryos generated in the study. **e**, Clonogenic formation of wild-type, *Fancd2*$^{-/-}$, *Polq*$^{-/-}$ and *Fancd2*$^{-/-}$*Polq*$^{-/-}$ MEFs with increasing concentrations of PARPi. **f**, A2780 cells were transduced with indicated shRNAs and xenotransplanted into both flanks of athymic nude mice. The tumour volumes for individual mice were measured biweekly for 8 weeks. Each group represents $n \geq 5$ tumours from $n \geq 5$ mice. **g**, Ki67 and γH2AX quantification in tumours treated with either vehicle or PARPi. **h**, Representative Ki67 and γH2AX staining of A2780-shFANCD2 xenografts expressing sh Scr or sh Polθ in athymic nude mice, treated with either vehicle or PARPi. Scale bars, 100 μM. **i**, *In vivo* competition assay design. **j**, Tumour chimaerism post-xenotransplantation for indicated conditions. **k**, Representative flow cytometry analysis of tumours before xenotransplantation (post-FACS sorting) or after xenotransplantation (post-transplant, PARPi). The percentage of GFP–RFP positive cells is indicated. **l**, Tumour chimaerism post-xenotransplantation for indicated conditions. For data in **j** and **l**, each circle represents data from one tumour and each group represents $n \geq 7$ tumours from $n \geq 6$ mice. Brackets show mean ± s.e.m. Data in **e**–**g** show mean ± s.e.m. For **f** each group represents $n \geq 6$ tumours from $n \geq 6$ mice.

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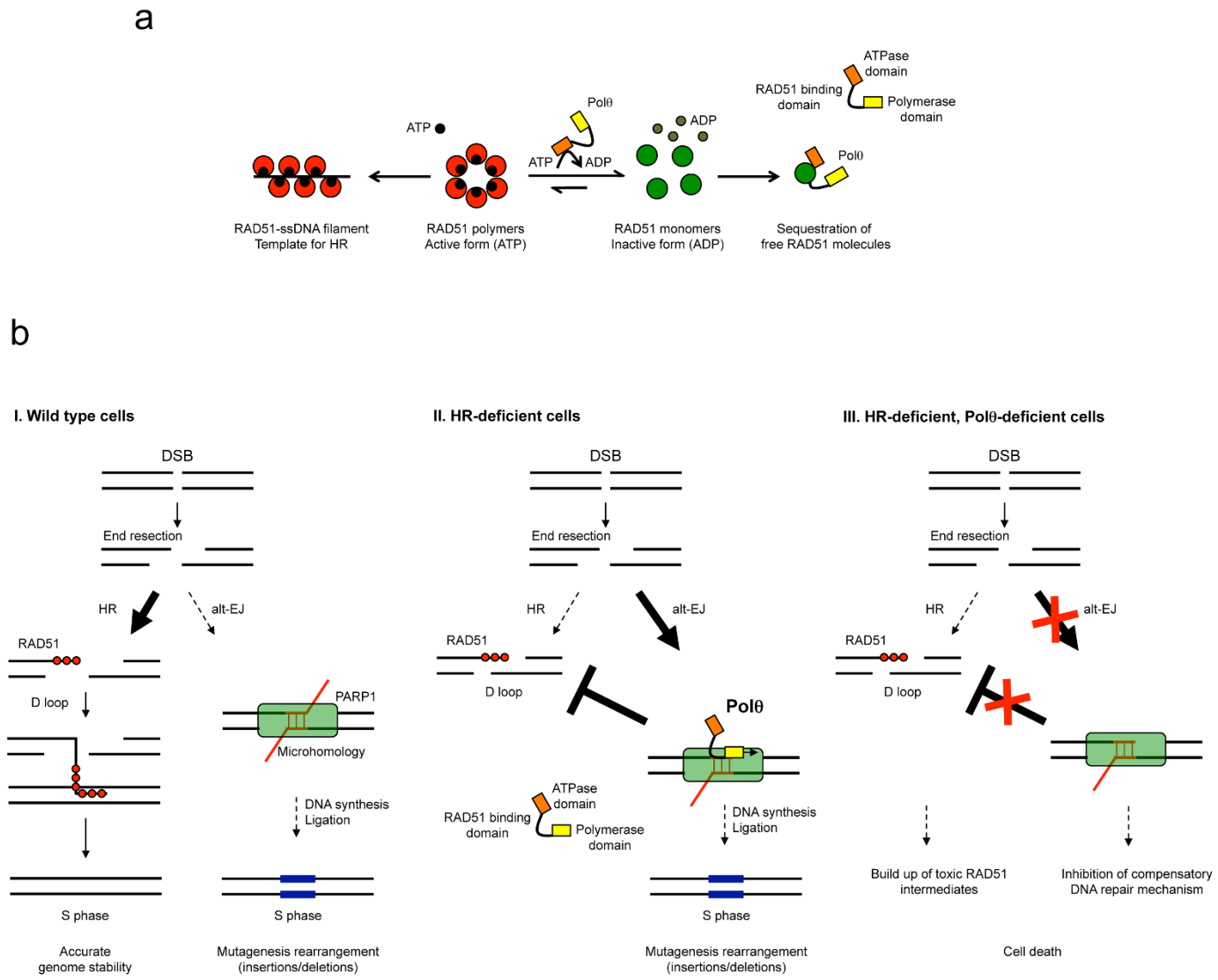
Extended Data Figure 8 | Polθ is required for HR-deficient cell survival and limits the formation of RAD51 structures in HR-deficient cells. **a**, Clonogenic formation of *Fancd2*$^{-/-}$*Polq*$^{-/-}$ MEFs transfected with full-length *POLQ* cDNA constructs in the presence of increasing concentrations of PARPi. **b**, Chromosome breakage analysis of FANCD2-depleted cells that were first transfected with the indicated siRNA and full-length *POLQ* cDNA constructs refractory to siPolθ1 and then exposed to MMC. **c**, DR-GFP assay in U2OS cells transfected with indicated siRNA. **d**, Quantification of baseline and IR-induced RAD51 foci in U2OS cells transfected with indicated siRNA. **e**, RAD51 recruitment to chromatin is enhanced by UV treatment. VU 423 cells (*BRCA2*$^{-/-}$) were collected at indicated time points after UV treatment and immunoblotting performed on the cytoplasmic, nuclear and chromatin fractions. **f**, RAD51 recruitment to chromatin in VU 423 cells (*BRCA2*$^{-/-}$) transfected with indicated siRNA. Histone H3 was used as a control for chromatin fractionation. All data show mean ± s.e.m.

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Extended Data Figure 9 | Polθ participates in error-prone DNA repair. **a**, End-joining reporter assay in U2OS cells transfected with indicated siRNA and/or treated with PARPi. **b**, End-joining reporter assay in U2OS cells transfected with indicated siRNA and *POLQ* cDNA constructs refractory to siPolθ1. **c**, UV damage-induced Polθ foci formation in U2OS cells. Polθ foci were abolished by pre-treatment with PARPi. UT, untreated. **d**, Mutation frequency was determined in damaged supF plasmid, recovered from siRNA-treated 293T cells. **e**, Non-synonymous mutation count in ovarian, uterine and breast TCGA. All data show mean ± s.e.m.

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Extended Data Figure 10 | Model depicting the role of Polθ in DNA repair. **a**, Mechanistic model for how Polθ limits RAD51–ssDNA filament assembly. According to this model, the ATPase domain of Polθ may prevent the assembly of RAD51 monomers into RAD51 polymers, perhaps by depleting local ATP concentrations. The RAD51 binding domains in the central region of Polθ may then sequester the RAD51 monomers, preventing filament assembly. **b,** (I) Under physiological conditions, Polθ expression is low and its impact on repair of DNA double-strand breaks (DSB) is limited. (II) When HR deficiency occurs, Polθ is then highly expressed and channels DSB repair towards alt-EJ. (III) In the case of an HR-defect, the loss of Polθ leads to cell death through the persistence of toxic RAD51 intermediates and inhibition of alt-EJ.

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